

---

# Electron Cyclotron Resonance Plasma Sources

---

*James E. Stevens*

## 1.0 INTRODUCTION

Electron cyclotron resonance (ECR) plasma sources are used for a variety of materials processing applications such as semiconductor etching and deposition. Their use in research became widespread in the late 1980's and ECR-based tools have now entered production. ECR plasma sources are termed *electrodeless* because they do not depend on current flowing to an electrode. ECR plasma sources are also wave-supported which allows the plasma to be generated remotely from tool surfaces. This chapter discusses the physics basis for the operation of typical ECR configurations such as the *divergent field* source pictured in Fig. 1.

ECR plasma sources were first investigated in the early 1960's. Miller, et al.<sup>[1][2]</sup> used essentially the configuration in Fig. 1 for space craft propulsion applications with the process chamber replaced by an opening for the plasma rocket exhaust. Plasma electrons created at high magnetic field are accelerated along the field lines toward the exhaust nozzle by the magnetic gradient force on the diamagnetic electrons. Ions are pulled along to maintain charge neutrality, thus producing a high energy, neutral exhaust

required for a spacecraft engine. Groups led by Consoli<sup>[3][4]</sup> and Dandl<sup>[5]</sup> carried out extensive research programs into ECR plasmas in connection with the controlled fusion effort. Studies of the physics of plasma waves near the electron cyclotron frequency were also carried out beginning in the 1960's.<sup>[6]–[14]</sup> These experiments confirmed many of the theoretical predictions for wave propagation and absorption.<sup>[15][16]</sup> ECR based ion sources are also used due to their high plasma density, low neutral pressure capability.<sup>[17]–[21]</sup>

Interest in ECR plasma sources increased tremendously after it was discovered that these sources could operate in new and desirable parameter regimes for semiconductor processing applications. The low pressure, high density regimes could not be achieved with other plasma tools at that time. Suzuki et al.<sup>[22]</sup> published a pioneering article utilizing ECR sources for semiconductor etching. ECR sources had several advantages over parallel plate “diode” tools, reactive ion etchers (RIE), and barrel etchers in common use in the 1970's–1980's, among them the ability to operate at much lower neutral gas pressures and much higher plasma densities while also maintaining lower plasma potentials. These advantages applied also to plasma-assisted deposition of silicon, silicon oxides, and metals<sup>[23][24]</sup> where good quality films could be achieved while maintaining low substrate temperatures. Most of the processing literature involving ECR sources is generic to other low pressure, high density sources, because ECR sources were the first of these types of sources to become widely available.

Much of the literature involving ECR plasma sources is a mixture of specific process applications along with brief descriptions of the ECR plasma properties. A review of the ECR processing literature is not attempted here. This chapter is, rather, a discussion of the physics of the ECR source itself with an emphasis on plasma topics which are important to materials processing needs. The literature cited is intended to illustrate the plasma properties of ECR tool operation rather than specific processing results. Previous review articles include those by Musil<sup>[25]</sup> on microwave plasmas, Holber<sup>[20]</sup> on ECR ion sources, Popov<sup>[26]</sup> on ECR sources for deposition, and Lieberman and Gottscho<sup>[27]</sup> on high density plasma sources. Section 2 of this chapter is a discussion of the basic physics and general operating principles of ECR plasma sources. Section 3 is a description of special configurations and applications of ECR sources. Section 4 is a discussion of some practical issues and open questions.

## 2.0 PRINCIPLES OF ECR SOURCE OPERATION

### 2.1 Plasma Properties of ECR Sources and Relation to Processing Needs

It is worth keeping in mind the reasons why ECR or other plasma sources are being studied for plasma processing. ECR sources operate in plasma regimes which have advantages over the older and more well-established parallel plate plasma processing tools. First, ECR sources operate at low neutral gas pressures, typically 0.2–10 mtorr, compared to 50–1000 mtorr for diode and RIE tools. Low pressure reduces ion collisionality in the substrate sheaths which is necessary for anisotropic etching of the smaller and increasingly high aspect ratio features in modern integrated circuits.

ECR sources also operate in the relatively high density range of  $n_i \approx 10^{11}$ – $10^{12}$  cm<sup>-3</sup>, compared to  $n_i \approx 10^{10}$  cm<sup>-3</sup> typical of RIE tools. Higher ion density implies higher ion-flux-driven processing rates ( $\Gamma_i \sim n_i$ ) which is important due to the high ownership costs of semiconductor processing tools. ECR sources, in this respect, fit well with the trend in the semiconductor industry towards single-wafer processing tools. High ion flux (typically  $\Gamma_{ie} \approx 10$  mA/cm<sup>2</sup>) is also advantageous for operating with nonpassivating chemistries which are being developed to exploit this new operating regime.

Another beneficial property of ECR plasma sources is low plasma potential, typically 15–30 eV without substrate biasing. Plasma etching is generally a synergistic process involving both physical and chemical surface reactions, and low plasma potential can increase the etching selectivity between two substrate materials by reducing the physical aspect of etching relative to the chemical. Low plasma potential also reduces physical damage to the substrate from high energy ion bombardment as well as reducing unwanted sputtering of materials from the remainder of the tool surfaces.

ECR tools can also provide independent control of the ion energy and ion flux. Ion energy is controlled by RF or DC biasing of the substrate holder while control of the ion and neutral flux is achieved by varying the microwave power and neutral gas pressure. ECR sources are wave-driven which has the unique feature of plasma formation remote from any physical surfaces. This is compared with inductively coupled sources where power is deposited within a *skin depth*,  $\delta \sim c/\omega_{pe} \sim 0.5$  cm/( $n_e \times 10^{-12}$  cm<sup>-3</sup>)<sup>1/2</sup>, of the plasma surface and capacitively coupled sources where power is

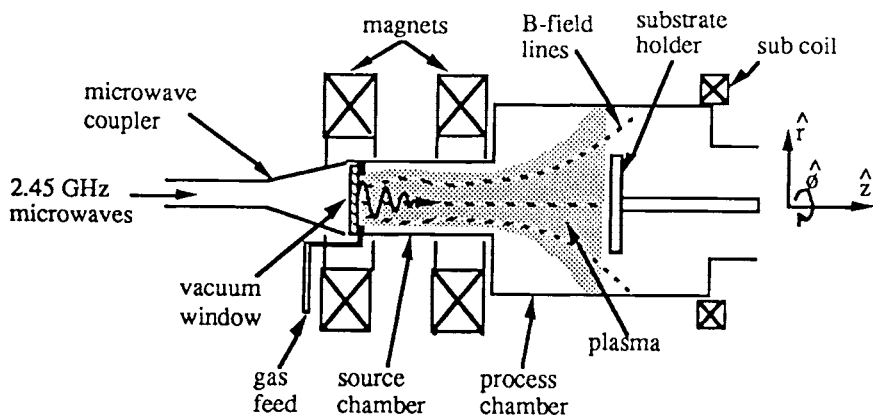
deposited within a few Debye lengths,  $d_{sheath} \sim (v_{sheath}/Te)^{3/4} \lambda_D \sim 0.01 \text{ cm}/(n_e \times 10^{-12} \text{ cm}^{-3})^{1/2}$ , of an electrode surface. In spite of being a wave-sustained source, the location of the microwave power absorption can be made quite localized in an ECR which can have significant processing advantages.

As with everything, there are disadvantages as well as advantages. ECR processing tools typically require  $\approx 1$  kilogauss level magnetic fields which imply extra expense for power supplies, cooling, power, and floor space. This is mitigated somewhat by the fact that the plasma source is a small component of the overall cost of the processing tool which can exceed \$1M. Microwave circuit components are somewhat more expensive and unfamiliar. High density operation implies a higher degree of molecular dissociation in ECR tools which is a problem for oxide etch. The higher density of neutral radicals can also lead to higher spontaneous, as opposed to ion-assisted, etch rates at the substrate surface. Plasma uniformity of ion and RF currents is a major concern for ECR tools due primarily to the presence of the magnetic field which limits cross field diffusion of the plasma. Process uniformity is usually directly related to plasma (ion flux) uniformity. Plasma uniformity is needed in etching in order to minimize the required over-etch of the material being removed and to reduce possible damage to underlying layers. Uniformity is also important in deposition applications because of the close tolerances on materials thickness and the fact that it is difficult to modify the thickness of a thin layer of material once it has been deposited. Plasma processing tools which overcome these problems and take advantage of the unique plasma parameters produced in ECR sources are now on the market.

## 2.2 Typical ECR Configuration and Description of Equipment

A brief description of the equipment for a divergent field ECR source will be given before delving into the physics of ECR source operation. A typical divergent field ECR plasma processing source is shown schematically in Fig. 1. ECR tools, because of their axisymmetric layout, are usually set up as a single-wafer configuration. Processing is done in a cylindrical vacuum chamber which is often divided into a *source region*, where most of the plasma production occurs, and a *process chamber* where the substrate is located. Vessel materials are typically stainless steel or aluminum and production tools always have insulating liners such as quartz or anodized aluminum to control metallic sputtering. A steady state magnetic field is

generated by one, two, or more electromagnets surrounding the source chamber. The magnetic field is typically a kilogauss or more in the source chamber and tapers off to tens or hundreds of gauss in the process chamber as the field lines flair to larger radii. Additional coils are sometimes placed around the process chamber to adjust the plasma density and uniformity.



**Figure 1.** Generic divergent-field ECR plasma processing tool.

Microwaves are introduced into one end of the source chamber through a dielectric vacuum window by a coupling structure. Quartz is the most common window material and alumina is also used in some cases. The microwave frequency is almost always 2.45 GHz because magnetron sources, similar to those used in everyday microwave ovens, are cheap and commercially available at that frequency. The microwave circuit (Fig. 2) typically consists of a magnetron microwave source, a ferrite circulator to protect the source by shunting reflected power into a dummy load, a reflected power monitor, tuning stubs, and the plasma coupler. These circuit elements are typically connected by  $3.8 \text{ cm} \times 7.6 \text{ cm}$  waveguide. Three tuning stubs allow any arbitrary plasma load to be matched and are located just before the coupler to minimize reactive power loss. Reflected microwave power is sampled through a directional coupler (typically  $\approx -50 \text{ dB}$ ) and converted to a DC signal by a diode detector. Most published ECR experiments to date

have used power levels between 0.1 and 1.5 kW although 5 kW systems are now commercially available at 2.45 GHz. Losses in the microwave circuit are usually negligible.

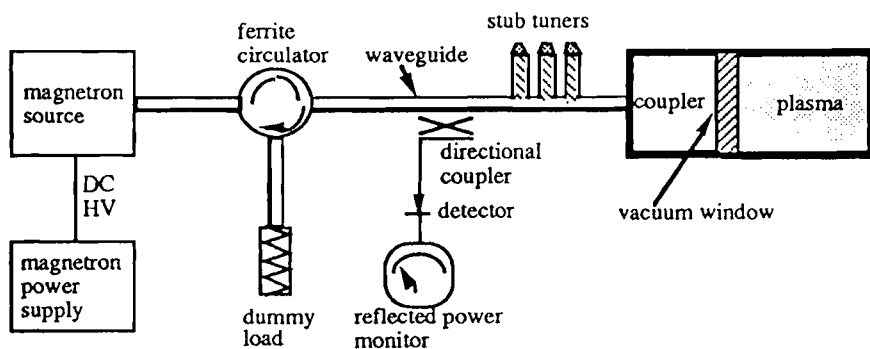


Figure 2. Microwave circuit for an ECR processing tool.

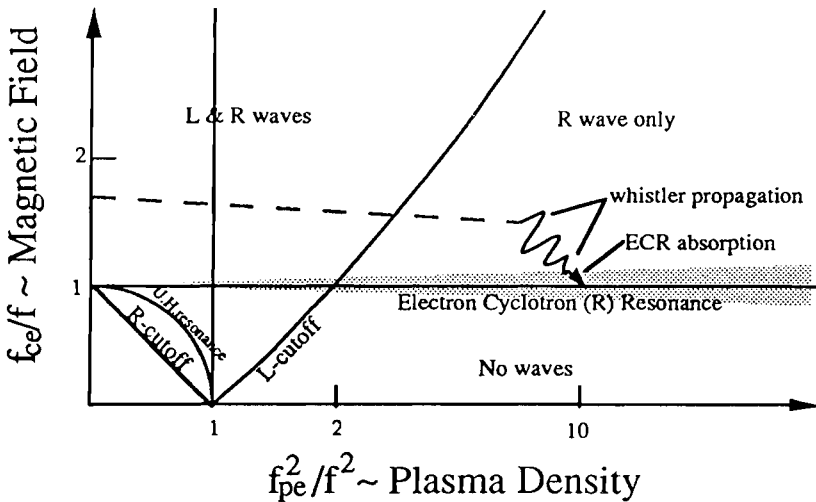
The microwave window is generally located under one set of coils near the point of highest magnetic field. The 2.45 GHz microwaves in the plasma are strongly absorbed near the ECR resonance magnetic field of  $B_{res} \approx 875$  G and propagate in the plasma at fields above  $B_{res}$ . Thus, a field higher than  $B_{res}$  is required in the source region at the vacuum window in order to allow the microwaves to propagate into the plasma before damping. Reactive gas at low pressure, typically below 10 mtorr, is dissociated and ionized by plasma electrons which absorb power at the electron cyclotron resonance. The plasma electrons are strongly magnetized and flow out of the chamber primarily along the field lines by diffusion and by the  $-\mu \nabla B$  force, where  $\mu$  is the magnetic moment given by  $\mu = \frac{1}{2} m_e v_{\perp}^2 / B$ ,  $v_{\perp}$  is the perpendicular electron velocity, and  $B$  is the magnetic field strength. Ions are forced along by the electric field set up by the charge separation and can also be lost perpendicular to the magnetic field due to their large cyclotron orbits.

Processing is done at the substrate (Fig. 1) by the ion and neutral flux produced in the ECR plasma. Gases containing fluorine, chlorine, or bromine are used for etching and are generally injected into the source chamber. A second gas distribution ring is often located downstream for

deposition applications when less dissociation of certain molecular species is desired. The maximum process rate is limited by the flow of reactants  $Q$  which is related to the operating pressure  $p$  and the pumping speed  $S$  by  $Q$  (sccm) =  $p$  (mtorr)  $S$  (liters/sec)/12.66.

### 2.3 Plasma Regimes of ECR Sources

The operation of ECR sources depends on propagating a high frequency wave through a magnetized plasma. The CMA diagram<sup>[15]</sup> shown in Fig. 3 parameterizes the possible plasma waves for a given frequency  $f$  versus density  $\sim (f_{pe}/f)^2$  (abscissa) and magnetic field  $\sim (f_{ce}/f)$  (ordinate). The plasma frequency is defined in CGS units as  $f_{pe} = 1/(2\pi)(4\pi n_e e^2/m_e)^{1/2} \approx 9000[n_e(\text{cm}^{-3})]^{1/2}$ , where  $n_e$  is the electron density,  $e$  is the electron charge, and  $m_e$  is the electron mass. The condition  $f = f_{pe}$  is represented by a vertical line in Fig. 3 and implies a critical density of  $n_e \approx n_c \approx 7.5 \times 10^{10} \text{ cm}^{-3}$  for  $f = 2.45 \text{ GHz}$ . The electron cyclotron resonance is represented by the horizontal line at  $f_{ce}/f = 1$ , where the electron cyclotron frequency  $f_{ce} = 1/(2\pi)(eB/m_e c) \approx 2.8 \text{ GHz/kG}$ ,  $B$  is the magnetic flux density, and  $c$  is the speed of light. The resonance magnetic field is  $B = 875 \text{ G}$  for  $f = 2.45 \text{ GHz}$ .



**Figure 3.** CMA diagram for plane waves showing parameter range for ECR plasma tools.

We are primarily interested in sources with densities well above  $n_c \approx 7.5 \times 10^{10} \text{ cm}^{-3}$ , i.e.,  $f_{pe} \ll f$ . Lower densities ( $n_e < 7.5 \times 10^{10} \text{ cm}^{-3}$ ) typically occur only in the plasma sheath or for very low power conditions in ECR processing sources. The assumption  $f_{pe} > f$  allows us to neglect many complicated effects discussed by Musil, et al.<sup>[8][28]</sup> and Popov<sup>[29]</sup> which occur near  $f \approx f_{pe}$  and greatly simplifies the analysis of ECR sources. The cold plasma dispersion relation appropriate for ECR sources is the Altar-Appleton-Hartree dispersion equation discussed by Stix.<sup>[15]</sup> For wave propagation primarily along the magnetic field, the cold plasma dispersion equation can be separated into a right hand polarized (RHP) mode

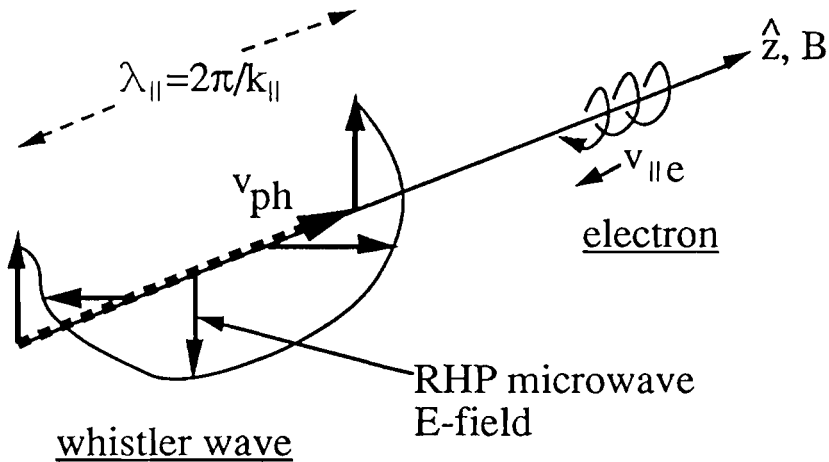
$$\text{Eq. (1)} \quad n^2 = R = 1 + f_{pe}^2/f^2 / [(f_{ce}/f)(k_{\parallel}/k) - 1]$$

and a left hand polarized (LHP) mode:

$$\text{Eq. (2)} \quad n^2 = L = 1 - f_{pe}^2/f^2 / [(f_{ce}/f)(k_{\parallel}/k) + 1]$$

where  $n = ck/\omega$  is the index of refraction of the wave,  $c$  is the speed of light,  $k = 2\pi/\lambda$  is the wavenumber,  $k_{\parallel} \approx k$  is the wavenumber parallel to the magnetic field,  $\omega = 2\pi f$  is the frequency in radians, and  $R$  and  $L$  are elements of the cold plasma dielectric tensor in Stix's notation. The RHP mode has a resonance ( $n \rightarrow \infty$ ) at the electron cyclotron frequency and power can be efficiently coupled to electrons at the resonance. Thus, the RHP mode is the relevant mode for ECR plasma sources. In addition, the RHP mode exists for arbitrarily high density provided that  $f_{ce}/f > 1$ , i.e., it has no cutoff. The RHP mode for  $f_{ce}/f > 1$  is often referred to as the *whistler mode*. A whistler in a bounded medium, usually with  $f \ll f_{ce}$ , is called the *helicon mode* and is used as the basis for another type of high density plasma source described by Boswell<sup>[30]</sup> and Chen<sup>[31]</sup>. The electron cyclotron resonance is pictured in Fig. 4, where the electric field of the RHP whistler wave is shown rotating around the magnetic field in the same sense as the plasma electrons and thus can remain in phase and accelerate electrons continuously when  $f \approx f_{ce}$ . Finally we note that  $f_{pe}^2/f^2 \gg 1$  for typical ECR densities while  $f_{ce}/f \approx 1$  near the ECR resonance, so that a wave refractive index of  $n \approx 5-10$  is typical. This implies that the plasma wavelengths,  $\lambda \approx \lambda_0/n \approx 1-2 \text{ cm}$ , are much shorter than the free space wavelength  $\lambda_0$ , and also small compared to the typical diameter of ECR sources ( $\approx 10-30 \text{ cm}$ ). Thus, the waveguide modes of the vacuum chamber have little relevance to the steady state situation but may be important for the initial breakdown of the plasma.





**Figure 4.** Resonance between a RHP whistler wave and an electron circling the magnetic field. The ECR resonance is Doppler (blue) shifted due to the finite parallel velocity of the electron.

The LHP mode is cut off ( $n < 0$ ) for densities above  $n_e = n_L \gtrsim 2n_c$ , as indicated by the line at  $\approx 45^\circ$  in Fig. 3. The LHP mode has no resonance in the parameter regime of interest and thus has no strong interaction with the plasma electrons in the low density regime where it propagates. However, the existence of the LHP mode can explain some of the anomalous behavior reported by Popov<sup>[32]</sup> and Carl, et al.,<sup>[33]</sup> where mode changes occur as the plasma density changes from  $n_e < n_L$  to  $n_e > n_L$ . Carl et al.<sup>[33]</sup> showed that standing wave patterns exist in the source when the LHP mode is present at low density ( $n_e < n_L$ ) while nearly complete absorption occurs at high density.

The path of microwave power flow in a typical ECR source between the window and the electron cyclotron resonance is indicated by the dashed and squiggly lines in Fig. 3. No plasma waves exist in the regime of high density and low  $B$ -field ( $f_{ce} < f$  and  $f_{pe}^2 > 2f^2$  in Fig. 3), so the magnetic field must be everywhere greater than  $B_{res} = 875 \text{ G}$  ( $f_{ce}/f > 1$ ) along the path of the wave in order for it to reach the ECR resonance. The dashed line

represents the fact that the plasma density jumps abruptly at the plasma sheath from  $n_e = 0$  to  $n_e \gg n_c$  over distances small compared to a plasma wavelength. Thus, plasma effects occurring in the neighborhood of  $f \approx f_{pe}$  can be neglected to first order. The squiggly line represents whistler wave propagation along a path of slowly increasing  $n_e$  and decreasing  $B$ , until absorption occurs near the electron cyclotron resonance on the so-called *magnetic beach*. A description of these processes is made more explicit in the following sections.

## 2.4 Microwave Coupling to the ECR Plasma

The most important concern in ECR coupler design is that it may influence the radial uniformity of the plasma density and the resulting uniformity of the process at the substrate. Another concern of coupler design is that of impedance matching between the plasma and the waveguide. Impedance matching in ECR sources is generally done by minimizing reflected power with adjustments of the tuning stubs (Fig. 2). Automatic tuning stubs are now commercially available. Once a plasma condition or a set of similar plasma conditions has been established, the tuner settings do not, in general, need to be changed much. However, impedance matching for multistep processes or processes where the microwaves are rapidly modulated could still be an issue. In addition, the microwave impedance can produce diagnostic information about the plasma density in the ECR source and contributes, in general, to our understanding of ECR source operation.

Many ECR coupler designs have been tried. Perhaps the simplest is to butt the end of a rectangular waveguide carrying the fundamental  $TE_{10}$  mode into the end of the cylindrical source vessel as reported by Popov.<sup>[34]</sup> This has the advantage of simplicity but the disadvantage of a non-axisymmetric electric field pattern. A better approach was to first flair the rectangular guide into a circular guide transforming the  $TE_{10}$  rectangular mode into the fundamental  $TE_{11}$  circular mode. Although the field patterns of the incident  $TE_{11}$  mode are not axisymmetric, it has been pointed out by Popov<sup>[35]</sup> that the plasma itself will decompose the incident  $TE_{11}$  mode into a propagating RHP  $TE_{11}$  mode and a reflected mode. The plasma thus selects an axisymmetric propagating RHP mode even though the incident mode is not axisymmetric. Note that this RHP mode selection occurs only when the plasma density at the window is sufficiently high to prevent the LHP mode from propagating, i.e., for  $n_e > n_L \approx 1.5 \times 10^{11} \text{cm}^{-3}$ .

Musil, et al.,<sup>[8]</sup> Vegas, et al.,<sup>[36]</sup> Fujiwara, et al.,<sup>[37]</sup> and Shindo, et al.<sup>[38]</sup> have utilized couplers which use a right hand polarized  $TE_{11}$  mode in cylindrical waveguide. Polarizers can be built in several ways according to Ragan,<sup>[39]</sup> the simplest by inserting a metal or dielectric slab of appropriate length at  $45^\circ$  to the direction of the E-field in the circular guide. The RHP- $TE_{11}$  mode is strictly 100% polarized only on axis and is less than 100% polarized off axis, but this is a small correction since the power flow is centrally peaked for this mode. The RHP coupler arrangement avoids the problem of trying to tune out reflected power from both the LHP and RHP modes at the same time which can cause hysteresis effects in the tuning behavior due to their different interaction with the plasma. However, reflected power from the window-plasma surface still occurs when launching the RHP mode alone since the plasma impedance does not necessarily match the waveguide impedance for that mode.

A further progression along the same line of reasoning is to use the dielectric window itself as an impedance matching element by making it a quarter-wave transformer between the RHP plasma wave and the waveguide mode. This configuration was explored twenty years ago in a basic plasma physics experiment by Musil et al.,<sup>[9]</sup> and is discussed in detail in the following paragraphs because it is simple to analyze and to implement.<sup>[18][40]</sup>

Microwave coupling in an ECR source can be modeled by assuming that the microwave circuit interacts with the plasma only at the plasma surface. Microwaves are thus assumed to propagate away from the window and be absorbed in the bulk plasma. Experimental data support these assumptions for the RHP mode. A plasma surface impedance can then be defined in terms of the density and magnetic field profiles within a fraction of a plasma wavelength ( $<1$  cm) of the plasma surface. Figure 5 shows the generic microwave circuit components and power flows in a divergent field ECR plasma tool. Microwave power  $P_f$  is applied from a source waveguide, through a tuning element, a connecting waveguide, and a dielectric window to a longitudinally magnetized plasma column. The tuning element acts to return the reflected power  $P_r$  to the plasma by transforming the combined impedance of all of the components to the right of it to match the source waveguide impedance. The one-dimensional model of microwave circuit for the RHP- $TE_{11}$  mode shown in Fig. 5 can be treated by standard transmission line theory. The plasma surface impedance  $Z_p$  is related to the load impedance  $Z_L$  at the outer surface of the window by the transmission line equation

$$\text{Eq. (3)} \quad Z_L = Z_W [Z_p + Z_W \tanh(\beta\ell)] / [Z_W + Z_p \tanh(\beta\ell)]$$

where  $\beta$  is the  $\text{TE}_{11}$  mode wavenumber for the window,  $\ell$  is the window thickness, and  $\tanh$  is the hyperbolic tangent. The window characteristic impedance for the  $\text{TE}_{11}$  mode,  $Z_W$ , is given by

$$\text{Eq. (4)} \quad Z_W = Z_0 / \sqrt{\epsilon_r - (f_c/f)^2}$$

where  $\epsilon_r$  is the dielectric constant for the window material ( $\epsilon_r \approx 3.8$  for quartz;  $\epsilon_r = 9.8$  for  $\text{Al}_2\text{O}_3$ ),  $f$  is the microwave frequency,  $f_c = 0.293 c/a$  is the  $\text{TE}_{11}$  waveguide cutoff frequency,  $c$  is the speed of light, and  $a$  is the waveguide radius, and  $Z_0 = 377 \Omega$  is the free space impedance. Choosing the window to be electrically a quarter-wavelength ( $\beta\ell = \pi/2$ ) gives the quarter-wave matching condition

$$\text{Eq. (5)} \quad Z_L = Z_W^2 / Z_p$$

An equation similar to Eq. (3) relates the impedance  $Z_L$  at the window to the impedance seen at the tuner,  $Z'_L$  (Fig. 5). The tuning circuit matches  $Z'_L$  to the source waveguide characteristic impedance  $Z_G$  in order to utilize all of the microwave power. The waveguide characteristic impedance  $Z_G$  for the cylindrical  $\text{TE}_{11}$  mode is

$$\text{Eq. (6)} \quad Z_G = Z_0 / \sqrt{1 - (f_c/f)^2}$$

The remaining unknown is the plasma surface impedance  $Z_p$ . The microwave fields in the plasma are the sum of many propagating and evanescent waveguide modes, all of whose fields must be matched to the waveguide fields at the plasma boundary in order to determine the plasma impedance. A first approximation is to consider the microwave fields in the plasma to be a plane wave (infinite plasma approximation). This works because the microwave electric and magnetic fields in the coupler, perpendicular to the chamber axis, are relatively uniform over distances comparable to a plasma wavelength ( $\lambda \approx 1\text{--}2$  cm). The plasma surface impedance  $Z_p$  then follows from Maxwell's equations for the microwave electric and magnetic fields,  $\bar{\mathbf{E}}$  and  $\bar{\mathbf{B}}$ ,

Eq. (7)  $\nabla \times \bar{\mathbf{E}} = -\partial \bar{\mathbf{B}} / \partial t$

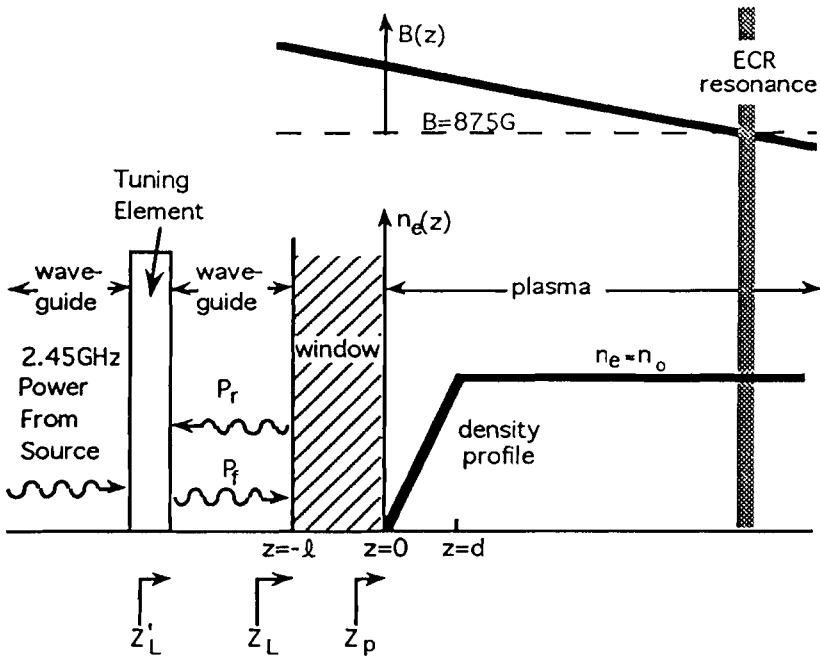
Eq. (8)  $\nabla \times \bar{\mathbf{B}} = c^2 \partial / \partial t (\bar{\bar{\mathbf{K}}} \bar{\mathbf{E}})$

where the cold plasma dielectric tensor  $\bar{\bar{\mathbf{K}}}$  is given by Stix

Eq. (9) 
$$\bar{\bar{\mathbf{K}}} = \begin{vmatrix} (R+L)/2 & -i(R-L)/2 & 0 \\ i(R-L)/2 & (R+L)/2 & 0 \\ 0 & 0 & P \end{vmatrix}$$

$R$  and  $L$  were defined by Eqs. (1) and (2), and

Eq. (10)  $P \approx 1 - f_{pe}^2 / f^2$



**Figure 5.** One dimensional ECR coupling model.<sup>[40]</sup>

The magnetic field is taken to be along the  $\hat{Z}$  direction (Fig. 1) and the window surface is in the  $\hat{X}$ - $\hat{Y}$  plane. It is reasonable to neglect  $\nabla_{\perp} \sim k_{\perp}$  relative to  $\nabla_{\parallel} \sim k_{\parallel}$  since the perpendicular wavenumber,  $n_{\perp}$ , is determined by the waveguide mode and is less than one while  $n_{\parallel}$  is determined by the plasma (Eq. 1) and is much larger than one. Perpendicular variations in the plasma density and magnetic field are also neglected. An  $\exp(-i\omega t)$  time dependence for the wave fields gives the condition for RHP as  $\mathbf{E}_{\mathbf{x}} = -i\mathbf{E}_{\mathbf{y}}$ , and results in the wave equation

$$\text{Eq. (11)} \quad \partial^2 \mathbf{E}_{\mathbf{x}} / \partial z^2 + R(z) k_0^2 \mathbf{E}_{\mathbf{x}} = 0$$

where  $k_0 = \omega/c$ . The plasma density, neglecting the sheath at the window, and the magnetic field are approximately constant over the  $<1$  cm coupling region, implying  $R(z) \approx \text{constant}$ . The solutions for  $\mathbf{E}_{\mathbf{x}}$  and  $\mathbf{B}_{\mathbf{y}}$  are then

$$\text{Eq. (12)} \quad \mathbf{E}_{\mathbf{x}} = E_0 \exp(ik_0 \sqrt{R} z)$$

$$\text{Eq. (13)} \quad c\mathbf{B}_{\mathbf{y}} = \sqrt{R} \mathbf{E}_{\mathbf{x}}$$

where  $E_0$  is a constant. The plasma surface impedance for the RHP wave is then

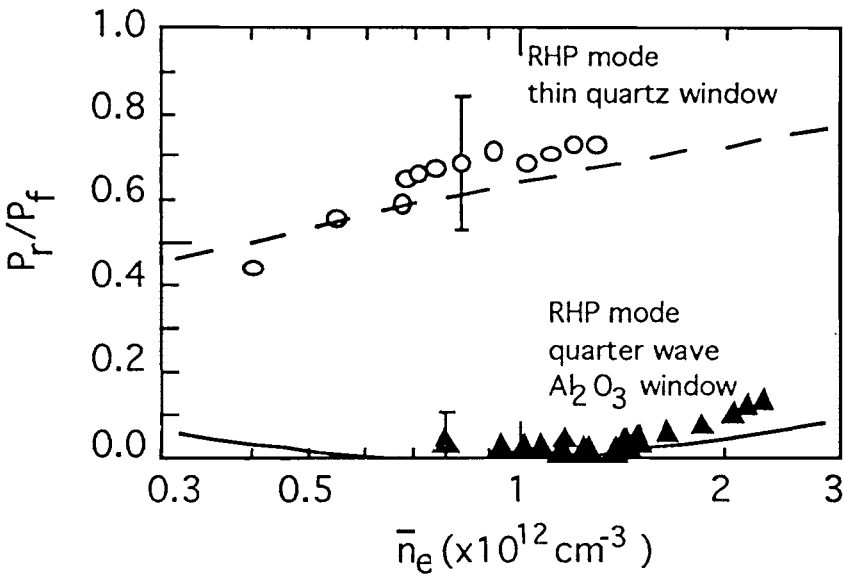
$$\text{Eq. (14)} \quad Z_p = Z_0 \mathbf{E}_{\mathbf{x}} / c\mathbf{B}_{\mathbf{y}} = Z_0 / \sqrt{R}$$

The plasma surface impedance from Eqs. 11–14 is real (resistive) if the sheath thickness is narrow compared to the plasma wavelength, i.e.,  $R(z) \approx \text{constant}$ . Reflected power at the window outside surface is given in terms of the forward power by

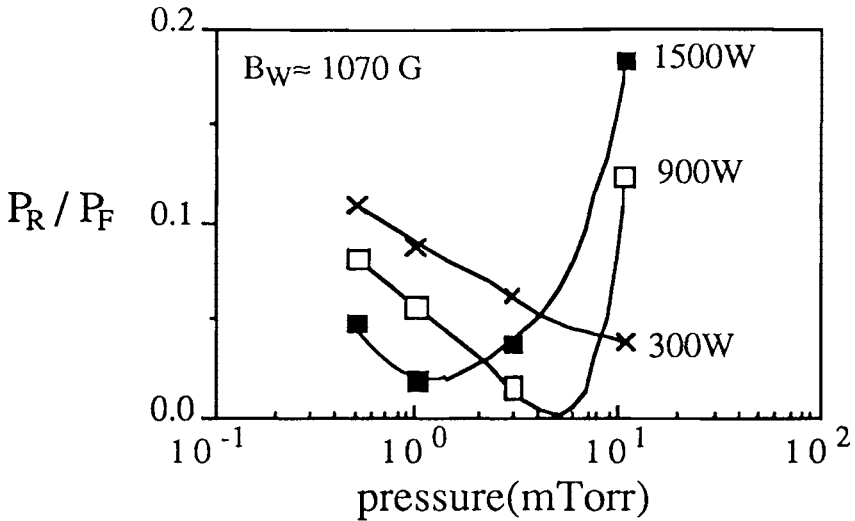
$$\text{Eq. (15)} \quad P_r/P_f = |(Z_L - Z_G)/(Z_L + Z_G)|^2$$

Measurements have verified the essential coupling behavior described above. Figure 6 shows the measured reflection coefficient,  $P_r/P_f$ , versus edge density for the RHP-TE<sub>11</sub> mode using both an electrically “thin” ( $\beta\ell < \pi/2$ ) quartz window (*open circles*) and a quarter-wave alumina window (*solid triangles*) with a window magnetic field of  $B_w = 950$  G. No external tuning was used for these measurements. The solid and dashed lines show the calculated reflection coefficients from Eqs. (6), (14), and (15) which

agree well with the measurements. Note that without tuning, the RHP reflected power for a thin window is quite high at the desired operating density of  $n_e \approx 10^{12} \text{ cm}^{-3}$ . Good matching ( $P_r/P_f < 5\%$ ) can be achieved for the quarter-wave window over an order of magnitude density range  $3 \times 10^{11} \gtrsim n_e \gtrsim 3 \times 10^{12}$  without tuning. The measured phase of the impedance for the quarter-wave window was also consistent with an almost purely real plasma impedance  $Z_p$  for a wide range of plasma densities.<sup>[40]</sup> Figure 7 shows that the reflection minimum shifts to higher pressure for lower microwave power because less power is required at high pressure to produce the optimum plasma density for coupling. Other ECR coupler designs should have the same qualitative behavior as Figs. 6 and 7 provided that the plasma density exceeds the LHP cutoff density and provided that the tuners are set to an optimum fixed location.



**Figure 6.** Microwave reflection of a  $\text{TE}_{11}$  RHP mode versus average source density for two microwave windows with  $B_w \approx 950 \text{ G}$  in argon.<sup>[40]</sup>



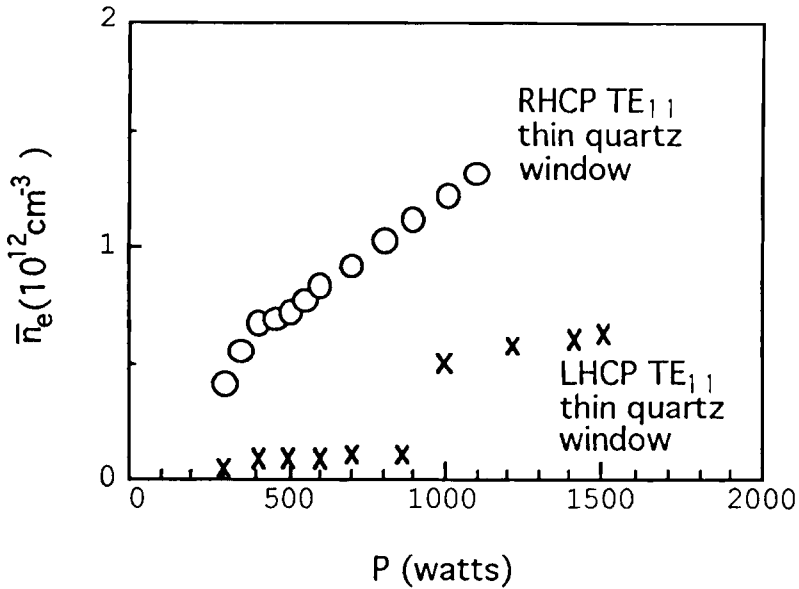
**Figure 7.** Coupling versus neutral pressure and microwave power ( $\approx$  density) in argon.

The effect of the coupler mode polarization is shown in Fig. 8. Data were taken in argon using a thin quartz window with no external tuning. The line average density in the ECR source region increases smoothly with power for the RHP-TE<sub>11</sub> mode. The LHP-TE<sub>11</sub> mode produces little density,  $\bar{n}_e \lesssim 1.2 \times 10^{11} \text{ cm}^{-3}$ , until the power exceeds  $\approx 1000 \text{ W}$ . The low source density is not unexpected because the LHP mode is cut off for densities  $n_e \lesssim 1.5 \times 10^{11} \text{ cm}^{-3}$  where  $L < 0$  (Eq. 2). The line average density in the source region jumps to  $\approx 6 \times 10^{11} \text{ cm}^{-3}$  for power greater than 1000 W and is correlated with the observation of nonlinear behavior in the frequency spectrum from a probe in the source region. However, the LHP mode produces plasma less efficiently than the RHP mode at all powers.<sup>[36][40][41]</sup>

Equations (1), (4)–(6), (14), and (15) indicate that it should be possible to obtain a good match using a RHP-TE<sub>11</sub> quarter-wave coupler for any ECR source density  $n_e$  by adjusting the window magnetic field,  $B_W$ . The window magnetic field which minimizes microwave reflection for a given plasma source density and window material is

$$\text{Eq. (16)} \quad B_W/B_{\text{res}} \approx 1 + (n_e/n_c)(\epsilon_r^{-2})[1 - (f_c/f)^2]$$



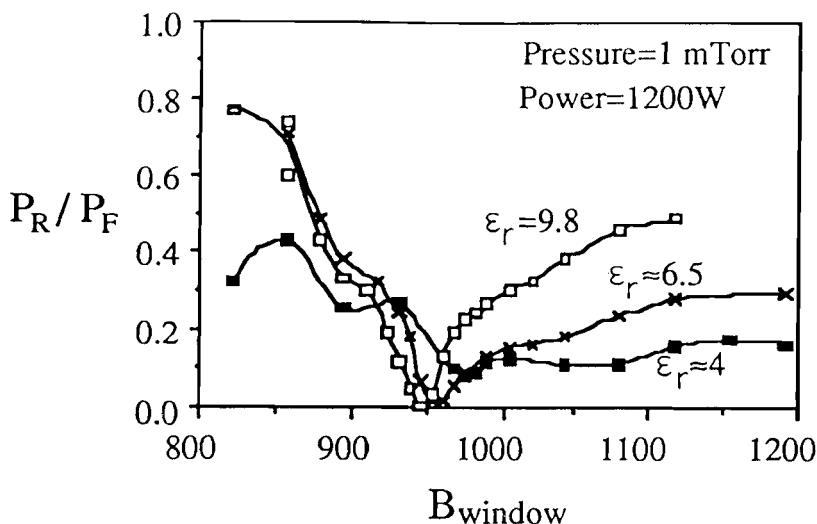


**Figure 8.** Plasma production versus microwave polarization in argon with  $B_0 \approx 950 \text{ G}$ .<sup>[40]</sup>

Figure 9 shows data of reflection versus magnetic field where the window dielectric constant is effectively varied by placing a second quarter-wave slab ( $\epsilon_{r2} < 2.1$ ) in back of an alumina quarter-wave window ( $\epsilon_{r1} = 9.8$ ), giving an effective window dielectric of  $\epsilon_r \approx \epsilon_{r1}/\epsilon_{r2}$  (apply Eq. 5 twice). As expected from Eq. (16), the minimum in reflection shifts to higher magnetic field with lower window dielectric constant.

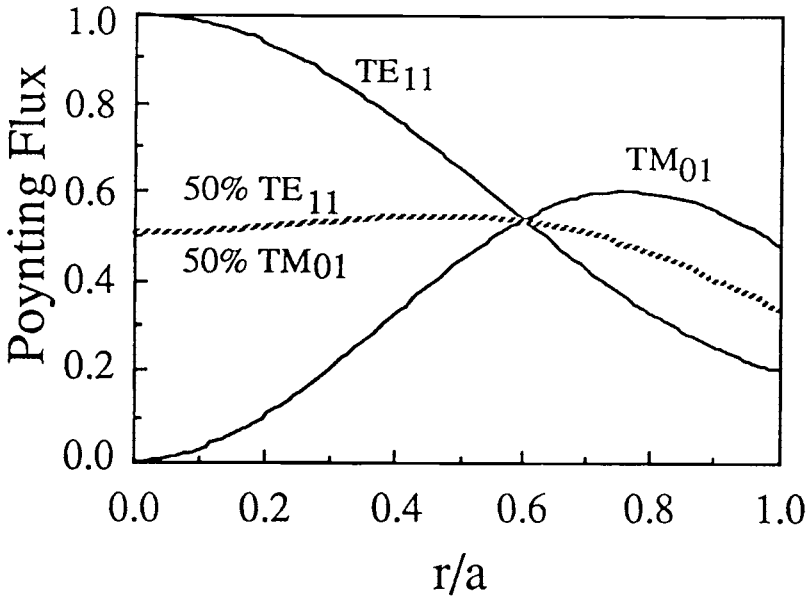
Other ECR coupler designs have been investigated in order to improve the uniformity of the plasma density profile. The radial distribution of power for the  $\text{TE}_{11}$  mode is given by the Poynting flux shown in Fig. 10. The  $\text{TE}_{11}$ -mode power is strongly peaked on axis and many authors have reported correspondingly-peaked plasma density profiles. The  $\text{TM}_{01}$  mode is the next highest cylindrical waveguide mode; it is also axisymmetric, and its Poynting flux is peaked off axis (Fig. 10). The  $\text{TM}_{01}$  mode has been utilized in ECR experiments by Popov<sup>[34]</sup> and Gorbatkin, et al.,<sup>[42]</sup> among others, and has often produced flatter or even hollow plasma profiles under various conditions. The  $\text{TM}_{01}$  mode also couples well over a wide parameter range after a three-stub tuner was set in one fixed location.<sup>[42]</sup> Other reports<sup>[40]</sup> of low power tuning difficulties for the  $\text{TM}_{01}$  mode were attributed to

a LHP wave component and disappeared once the plasma density exceeded  $n_L \approx 2n_c \approx 1.5 \times 10^{11} \text{ cm}^{-3}$ . Although the  $\text{TM}_{01}$  mode is unpolarized in the external waveguide, it can still be decomposed locally at the plasma surface into plane polarized waves due to the short plasma wavelengths. This picture has the microwaves being emitted into the plasma as local plane waves from the window-plasma surface with a polarization and power flux given by the coupler waveguide mode. This approximate decomposition into plasma plane waves is used in the next section as the initial condition for wave propagation in the ECR source. A supporting argument for this local decomposition is that, at best, only 50% of the power can be coupled without tuning using a  $\text{TM}_{01}$  or linearly-polarized  $\text{TE}_{11}$  mode through a quarter-wave matching window, whereas almost 100% can be coupled using a RHP  $\text{TE}_{11}$  waveguide mode. ECR couplers described by Matsuo, et al.<sup>[43]</sup> and Takahashi, et al.<sup>[44]</sup> have utilized dielectric slabs, Fukuda, et al.<sup>[45]</sup> reported utilizing tapered or rounded bell jars, and Nishimura, et al.<sup>[46]</sup> utilized concave window shapes for more uniform introduction of the microwaves.



**Figure 9.** Coupling versus window magnetic field with several effective window dielectric constants.

The previous discussion of ECR coupling assumed that the plasma already exists in a steady state. The initial plasma formation (breakdown) depends on the vacuum microwave fields. The vacuum chamber will act like an empty microwave cavity without the plasma load present. Breakdown in ECR sources can sometimes be difficult if the  $B_{res} = 875$  G resonance surface location is near the null of the electric field standing wave pattern or if the coupler is poorly matched to the empty vacuum chamber. Breakdown can be triggered by temporarily sweeping the window magnetic field through a value of  $B_{res}$ .



**Figure 10.** Radial profile of Poynting flux for TE<sub>11</sub> and TM<sub>01</sub> cylindrical waveguide modes.<sup>[50]</sup>

## 2.5 Wave Propagation and Power Absorption in ECR Sources

Transport of the microwave power is important in ECR sources because it affects the uniformity of the plasma density and thus of the

process. Wave deposition is especially important in strongly magnetized sources such as ECRs because plasma transport across the magnetic field is limited. Microwave propagation in an ECR plasma source depends on both the plasma density and magnetic field according to Eq. (1), and the vacuum chamber walls also influence the propagation through the wavenumber selection. The effect of plasma density on the propagation of whistler waves was studied long ago by Smith, et al.<sup>[47]</sup> in connection with experiments in the ionosphere, where it was noted that whistler waves can be ducted along the magnetic field lines in plasma density troughs. Laboratory measurements of the wave fields confirmed the whistler dispersion equation (Eq. 1) as well as collisionless electron cyclotron damping by McVey, et al.<sup>[10]</sup> and ion cyclotron damping by Uman, et al.<sup>[48]</sup>. Later laboratory experiments by Stenzel<sup>[12]</sup> and Sugai, et al.<sup>[14]</sup> verified whistler wave ducting in density troughs. The same wave propagation and damping behavior was seen in higher power experiments by Tanaka, et al.,<sup>[41]</sup> Quon, et al.,<sup>[49]</sup> and Stevens, et al.<sup>[50]</sup> where the ECR waves produced the plasma.

An example of RHP microwave propagation and absorption is shown by interferograms in Figs. 11a,b. The plasma wave was measured along the axis of an ECR etch tool using a coaxial RF probe with the probe tip bent perpendicular to the axis. The probe signal was heterodyned with a reference signal taken from the microwave forward power giving an output proportional to  $E_0 \sin(\phi)$  where  $E_0$  is the amplitude of the microwave electric field and  $\phi$  is an arbitrary phase angle. The interferograms were taken with the magnetic fields shown in Fig. 11c. Note that the microwaves propagate for  $B \gg B_{res}$  and damp for  $B \lesssim 1.1 B_{res}$ .

Analytic solutions for the microwave fields in plasma-filled waveguides have been given by Allis, et al.<sup>[16]</sup> for special cases where the plasma density and magnetic fields are uniform. However, a realistic ECR source geometry has nonuniform density and magnetic fields in at least the  $\hat{r}$  and  $\hat{z}$  (= source axis) coordinates. Thus, an exact solution to the propagation problem requires at least a two-dimensional numerical solution, not to mention time-dependent or azimuthal plasma density variations. Such a complete model for wave propagation in an ECR plasma has not yet been reported. Fortunately, a good approximation for the microwave power flow in an ECR source can be obtained by the method of ray tracing. The ray (or WKB, or plane wave) approximation is valid as long as the wavelengths in the plasma, typically  $\lambda \approx 1\text{--}2$  cm, are short compared to the density gradient scale lengths in the plasma. Ray tracing can easily account for wave

refraction due to nonuniform plasma density and magnetic fields and for wave reflections from the vacuum chamber walls, and it thus gives us insight into the wave-plasma dynamics. The ray equations are given by:

$$\text{Eq. (17)} \quad d\bar{r}/dt = \frac{-\partial D/\partial \bar{k}}{\partial D/\partial \omega} = v_g$$

$$\text{Eq. (18)} \quad d\bar{k}/dt = \frac{-\partial D/\partial \bar{r}}{\partial D/\partial \omega}$$

where  $D = n^2 - R = 0$  is the dispersion equation given in Eq. (1) and  $\bar{r}$ ,  $\bar{k}$  are the position and wavenumber vectors. Equation (17) determines the group velocity,  $v_g$ , i.e., the trajectory of power flow in the plasma. The value of  $D$  depends on density and magnetic field and is a function of the  $\hat{r}$  and  $\hat{z}$  coordinates in the axisymmetric system shown in Fig. 1.

The initial power  $P_0$  carried by a ray is determined by the local Poynting flux of the coupler mode (see Fig. 10) where use is made of the local emission approximation discussed in the previous section. The power absorbed in the plasma along the ray path is given by

$$\text{Eq. (19)} \quad P_{abs}(z) = P_0 [1 - \exp(-\int 2k_{\parallel i} dz)]$$

where the integral is taken along the ray path beginning at the window and  $k_{\parallel i}$  is a small imaginary part of the wavenumber  $\bar{k} = \bar{k}_r + i\bar{k}_i$  which is determined by a Taylor expansion for the dispersion equation  $D \approx D_r + iD_i + i\bar{k}_i \partial D/\partial k$ , where  $D_r \approx 0$  is the real part of  $D$ . The hot plasma dispersion equation for RHP whistler waves assuming a Maxwellian electron distribution function gives  $D_i$  approximately as:

$$\text{Eq. (20)} \quad D_i \approx \omega_{pe}^2 / (\omega k_{\parallel} v_{te}) \sqrt{\pi} \exp \left\{ - \left[ (\omega - \omega_{ce}) / k_{\parallel} v_{te} \right]^2 \right\}$$

where  $v_{te} = (2k_B T_e / m_e)^{1/2}$  is the electron thermal velocity and  $k_B$  is the Boltzmann constant. Assuming  $\bar{k} \approx k_{\parallel}$  gives an expression for the collisionless wave absorption coefficient as

Eq. (21)

$$k_{\parallel i} \approx -D_i / \partial D / \partial k_{\parallel} \approx \omega \omega_{pe}^2 / (2c^2 k_{\parallel}^2 v_{te}) \sqrt{\pi} \exp \left\{ - \left[ (\omega - \omega_{ce}) / k_{\parallel} v_{te} \right]^2 \right\}$$

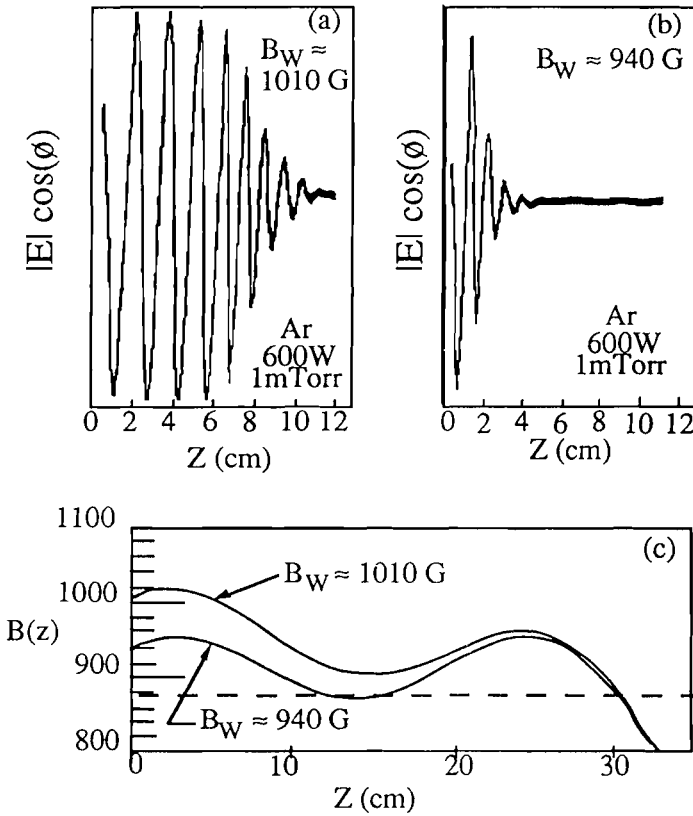


Figure 11. Wave interferograms for two magnetic fields.<sup>[50]</sup>

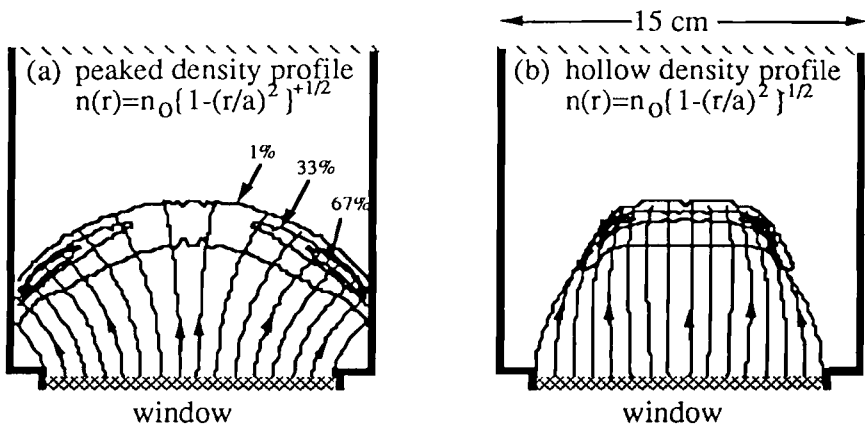
Figure 12 shows how whistler ray trajectories propagate and damp for two assumed density profiles in an ECR source, one profile peaked on axis and one hollow. It can be seen that the microwave power flow responds strongly to the plasma density profile. Equation (21) indicates that wave absorption is stronger at higher densities and higher temperatures, and thus damping occurs sooner for cases with higher powers and higher pressures as the wave approaches the ECR resonance field. Fidone<sup>[51]</sup> has shown that Eqs. (1) and (21) predict damping rates in error by as much as a factor of two in the regime  $|v_{||}| < 2v_e$  where most of the ECR damping actually takes place. This typically shifts the predicted damping location by less than a centimeter axially. A more accurate ray damping rate requires the full hot plasma dispersion equation, which for  $\vec{k} = k_{||}$  is given by

$$\text{Eq. (22)} \quad D \approx n_{||}^2 - 1 - \omega_{pe}^2 / (\omega k_{||} v_{te}) Z[(\omega - \omega_{ce}) / k_{||} v_{te}]$$

where  $Z$  is the plasma dispersion function for a Maxwellian distribution.<sup>[52]</sup> Wave absorption occurs where the Doppler-broadened resonance condition

$$\text{Eq. (23)} \quad \omega - \omega_{ce} - k_{||} v_{||} \approx 0$$

is satisfied. Strong absorption occurs for  $v_{||} \lesssim 2v_{te}$  in typical ECR plasma sources. Damping due to collisions can also be included<sup>[15][16]</sup> by replacing all occurrences of  $m_e$  by  $m_e(1 + i\nu_c/\omega)$  in Eq. (22), where  $\nu_c$  is the average electron collision frequency.



**Figure 12.** Ray trajectories and contours of relative power deposition in an ECR source for (a) a parabolic peaked radial density profile and (b) a hollow density profile.<sup>[50]</sup>

The interferograms shown in Fig. 11a–b are consistent with the Doppler-broadened ECR condition, which implies strong absorption for magnetic fields in the range  $810 \lesssim B \lesssim 940$  G for  $n_e \approx 10^{12} \text{ cm}^{-3}$  and  $T_e \approx 6$  eV. The actual magnetic field where absorption occurs depends on the details of how the microwave power reaches the absorption layer according to Eq. (19). Absorption typically occurs around  $B \approx 950 \pm 25$  G ( $v_{\parallel} \lesssim 2v_{te}$ ) for cases like that of Fig. 11a ( $B_{win} \approx 1010$  G) where the waves approach from a high magnetic field. This field is  $\approx 10\%$  above the  $B_{res} \approx 875$  G resonance condition where  $\omega \approx \omega_{ce}$  for 2.45 GHz. Equation (23) shows that it is actually fast electrons moving toward the wave (and towards the window) which absorb the microwave power, since the wave exists for  $\omega - \omega_{ce}$  negative which implies  $v_{e\parallel}$  is negative (see Fig. 4). Neglecting collisions, these tail electrons will be reflected by a magnetic mirror before hitting the window provided that  $B_{win}/B_o > v_o^2/v_{\perp o}^2$ , where  $v_{\perp o}$  and  $v_o$  are the perpendicular and total electron velocity evaluated at the absorption location.

The ray equations predict certain consequences for whistler waves in ECR sources. First, propagation is initially along the magnetic field lines. Using Eqs. (1), (17)–(18) gives the ratio of perpendicular to parallel group velocities as

$$\text{Eq. (24)} \quad v_{g\perp}/v_{g\parallel} \approx k_{\perp}/k_{\parallel}(\beta - 2)/[2(\beta - 1) + \beta(k_{\perp}/k_{\parallel})^2]$$

where  $\beta = \omega_{ce}k_{\parallel}/\omega k \approx \omega_{ce}/\omega$  and  $1 < \beta < 2$  for ECR sources. The ray equations imply that initially  $v_{g\parallel 0} \gg v_{g\perp 0}$ , because initially  $k_{\parallel} \gg k_{\perp}$ , where  $k_{\perp}$  is set by the coupler mode,  $k_{\parallel}$  is set by the plasma, and  $\beta$  is typically greater than 1.2. A consequence of this is that the microwaves from a small ECR source region can be spread out if the magnetic field lines flair out rapidly from the window. This is utilized in some ECR plasma tools with strongly diverging fields in the source.<sup>[46]</sup> Near the ECR resonance ( $\beta \rightarrow 1$ )  $v_{g\perp}$  can become comparable to  $v_{g\parallel}$ . The ray equations also imply that rays propagating along the magnetic field are refracted towards regions of lower density in the ECR source. This can be shown by starting with initial conditions  $k = k_{\parallel 0}$ ,  $k_{x0} = 0$ , and  $\omega_{pe}^2 \approx \omega_{pe0}^2(1 + x/\ell)$  where  $\ell$  is a radial density gradient scale length. Applying Eq. (1) in Eqs. (17)–(18) shows that  $k_x$  becomes positive and  $v_{gx}$  takes the same sign as  $(\omega_{ce}k_{\parallel}/\omega k - 2)$ , as the ray begins to propagate. This implies that the ray bends towards lower density for  $\omega_{ce}/\omega \lesssim 2$  which is the case in ECR sources. Figure 12 illustrates this effect for an ECR source. Waves refract towards regions of high density<sup>[47]</sup> in the opposite regime ( $\omega_{ce}k_{\parallel}/\omega k > 2$ ) which applies to helicon sources.

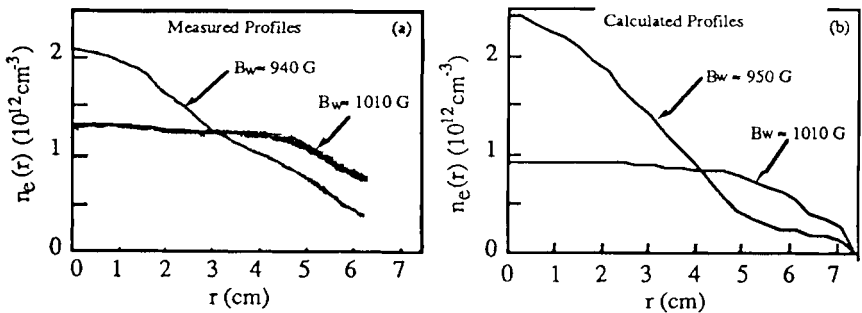


The discussion in this section has focused on how the microwave power transport in an ECR source depends on plasma density. Conversely, the plasma density and density profile in a strongly magnetized ECR source depends on the microwave power transport. Popov et al.<sup>[35]</sup> and Gorbalkin, et al.<sup>[42]</sup> have observed flat, peaked, and hollow modes in ECR sources for various reactor conditions. Radially uniform “flat” density profiles are desired in processing applications. Tulle,<sup>[11]</sup> Popov,<sup>[35]</sup> Samukawa, et al.,<sup>[53]</sup> and Tanaka, et al.<sup>[54]</sup> have noted that radially uniform density profiles occurred in ECR sources when the window magnetic field was raised well above the resonance field implying that the microwave power propagates some distance ( $\geq 10$  cm) before damping. This profile flattening effect has been attributed to the self-consistent interaction between wave propagation and the plasma density by Stevens, et al.<sup>[50]</sup> Improved plasma uniformity occurs because the microwave power propagates away from regions of high plasma density before damping. Figure 13a shows examples of radial density profiles in the source region for the two magnetic field conditions of Fig. 11. The density profile for a window magnetic field  $B_W \approx 940$  G is highly peaked while the profile for the case of  $B_W \approx 1010$  G is relatively flat over the inner 2/3 of the 7.5 cm radius source chamber. The two profiles in Fig. 13a can be explained in terms of the location of the microwave power deposition shown in Fig. 11. The wave is absorbed close to the window for  $B_W \approx 940$  G because the Doppler-broadened ECR resonance condition is already satisfied there (Eq. 23). This causes the power deposition profile in the plasma to maintain the peaked radial power distribution of the TE<sub>11</sub> mode (Fig. 10), producing a correspondingly peaked density profile (we neglect radial plasma losses for this discussion). The relatively flat profile for the  $B_W \approx 1010$  G case is a result of a spreading of the microwave power due to wave refraction by the radial plasma-density profile.

A simple model for the self-consistent wave-plasma interaction in an ECR source can predict the observed peaked and flat density profiles. The model assumes that the plasma density created by local microwave deposition is confined along those same magnetic field lines. For simplicity, diffusion across the field lines is assumed to be zero and diffusion along the magnetic field lines is replaced by a time constant  $\tau_e$ . The electron density,  $n_e(r)$ , thus satisfies the continuity equation

$$\text{Eq. (25)} \quad \frac{dn_e(r)}{dt} + n_e(r)/\tau_e = S(r)$$

where  $S(r)$  is the ionization rate due to the localized microwave energy deposition. The source term of  $S(r) \approx (\text{ray power/volume}) \times (1 \text{ particle}/60 \text{ eV})$  is chosen to give densities similar to Fig. 13 at 1 mtorr pressure. The density at any particular time and radius decays on a time scale of  $t_e \approx L/c_s \approx 1.4 \times 10^{-4} \text{ sec}$ , where  $L$  is the source length,  $c_s = (k_B T_e/M_i)^{1/2}$  is the sound speed, and  $M_i$  is the ion mass. Argon ( $M_i/M_p = 40$ ) is taken for the gas, where  $M_p$  is the proton mass, and the measured source electron temperature of  $T_e \approx 6 \text{ eV}$  is used. Equation (25) is solved by alternately calculating the local electron ionization rate  $S(r)$  from Eqs. (17)–(21) and then updating the density from Eq. (25) using a time step,  $dt$ , which is short compared to the plasma density decay time, i.e.,  $dt \lesssim 0.01 \tau_e \approx 10^{-6} \text{ sec}$ . The waves traverse the plasma on an even shorter time scale of  $L/v_g \approx 10^{-8} \text{ sec}$ . The calculated density profiles shown in Fig. 13b is an average over many confinement times ( $t_{ave} \approx t_e$ ). The calculated profiles in Fig. 13b show qualitatively the experimentally observed peaked and flat profiles of Fig. 13a. The  $B_w \approx 940 \text{ G}$  case, where the wave absorbs close to the window, shows the expected peaked density profile because the wave power does not have a chance to spread out. The density profile for the  $B_w \approx 1010 \text{ G}$  case is predicted to be relatively flat with a gentle  $\approx 10$ – $20\%$  decrease in average density over the inner  $2/3$  of the source region. This small gradient allows the initially peaked power flux to spread out. An interesting prediction of the model is that the ray trajectories constantly refract away from regions of high density and never converge. This effect leads to the prediction that the plasma density profiles fluctuate on time scales  $t \lesssim \tau_e$ , and the rays tend to bunch up locally depending on the density profile at a particular time.



**Figure 13.** (a) Measured plasma density profiles versus magnetic field using 1 mtorr argon and 600 W. (b) Calculated plasma density profiles.<sup>[50]</sup>

Large density fluctuations were observed by Stevens, et al.<sup>[55]</sup> in ECR sources on 100  $\mu$ s time scales and an inverse correlation between the local plasma density and the local microwave power density has been shown. These density and microwave fluctuations are consistent with the model predictions but it has not been proven that they are the same effect. The simple model described above shows that more realistic ECR plasma simulations may have to deal with the time-dependent behavior of the discharge. ECR models of the microwave propagation using either ray tracing or a full numerical model for the microwave fields are now being combined with more realistic plasma models.

Experiments combining the  $TM_{01}$  and  $TE_{11}$  modes on the same source<sup>[55]</sup> shown that the initial coupler mode matters most when the microwaves damp quickly near the window, while the coupler mode has little effect when the microwaves propagate a number of wavelengths. This shows that the microwaves indeed respond to the plasma and that wave propagation can mitigate the initial conditions to some extent. Even in “closely coupled” ECR tools such as those of Fukuda, et al.<sup>[56]</sup> and Samukawa, et al.,<sup>[57]</sup> where the etching is done in the source itself, the microwaves are allowed to propagate many plasma wavelengths ( $\geq 10$  cm or  $\geq 5\lambda$ ) before ECR damping which allows wave refraction to spread out the microwave power radially.

## 2.6 Particle and Energy Balance in ECR Plasma Sources

Calculations of the global (zero-dimensional) plasma parameters from the external control parameters have been made by Dandl and Guest<sup>[58]</sup> for ECR sources and by Lieberman and Gottscho<sup>[27]</sup> for low pressure tools in general. These calculations are useful for establishing general trends in the plasma and chemistry versus various parameters. The electron temperature is first determined from the steady state electron continuity equation:

$$\text{Eq. (26)} \quad dn_e/dt = 0 = n_e n_n \langle \sigma v_e \rangle_i - n_e n_i \langle \sigma v_e \rangle_r - n_e / \tau_e$$

where  $\langle \sigma v_e \rangle_i$  and  $\langle \sigma v_e \rangle_r$  are the electron velocity times the ionization and recombination cross sections for the particular gas averaged over the electron distribution,  $\tau_e$  is the electron particle confinement time, and the neutral density  $n_n$  is related to the pressure at room temperature by

$$\text{Eq. (27)} \quad n_n = p \text{ (mtorr)} \times 3.5 \times 10^{13} \text{ cm}^{-3}$$

The electron and ion particle confinement times are equal because the plasma is electrically neutral over distances larger than a Debye length. The plasma ions are lost to the walls along the magnetic field at roughly the sound speed,  $c_s = (eT_e/M_i)^{1/2}$ , if one neglects ion drifts set up by internal variations of the plasma potential which can exist for distances less than an ion mean free path. Thus,  $\tau_e = \tau_i \approx L/c_s$ , where  $L$  is the source length along the magnetic field direction. Neglecting recombination, the electron temperature is determined from the neutral pressure and gas species by solving

$$\text{Eq. (28)} \quad p \text{ (mtorr)} \approx c_s / (L \langle \sigma v_e \rangle_i 3.5 \times 10^{13})$$

This function is plotted in Fig. 14a assuming  $L = 20$  and  $40$  cm and using the ionization cross sections for argon.<sup>[59][60]</sup> It is seen that the electron temperature is relatively insensitive to pressure in the mtorr pressure range but increases strongly at sub-millitorr pressures. Significant recombination, electron attachment, or radial losses will shorten the effective length  $L$  and shift the curves up.

The electron density is determined by the steady state energy balance equation given by

$$\text{Eq. (29)} \quad d(3/2 n_e e T_e) / dt = 0$$

$$= P_\mu / V - n_e n_n e (E_i \langle \sigma v_e \rangle_i + E_x \langle \sigma v_e \rangle_x + 3 T_e m_e / M_i \langle \sigma v_e \rangle_{el}) - (3/2 n_e e T_e) / \tau_E$$

where  $P_\mu$  is the microwave power input in watts,  $V$  is the plasma volume in  $\text{cm}^3$ ,  $n_e$  and  $n_n$  are in  $\text{cm}^{-3}$ ,  $e = 1.6 \times 10^{-19}$  coulomb is the unit charge,  $\langle \sigma v_e \rangle_x$  is the electron velocity times the excitation cross section averaged over the electron distribution in  $\text{cm}^3/\text{s}$ ,  $E_i$  and  $E_x$  are the ionization and excitation energies in eV,  $\langle \sigma v_e \rangle_{el}$  is the average electron-neutral elastic collision cross section,<sup>[60]</sup> and the last term is the energy lost in an electron energy confinement time  $\tau_E$  due to transport. The elastic energy loss term can be neglected in the operating regime of most ECR sources where  $p < 10$  mtorr. The last term is determined by the ion behavior at the sheaths. In general, the plasma will take on a positive plasma potential  $V_p$  given by

$$\text{Eq. (30)} \quad V_p \approx V_f + T_e / 2 \ln(M_i / 2.3 m_e)$$

where the floating potential  $V_f = 0$  is assumed for the chamber walls. Electrons lost at the sheath must have an energy of at least  $V_p$  in order to overcome the sheath potential ( $=V_p - V_f$ ). The average energy of electrons lost at the sheath is  $E_e \approx V_p + 2T_e$  for a Maxwellian distribution, of which  $V_p$  eV is transferred to the ions falling through the sheath. Thus,

$$\text{Eq. (31)} \quad (3/2T_e)/\tau_E \approx (V_p + 2T_e)/\tau_e$$

The resulting equation for the electron density in terms of microwave power, source size, gas, and electron temperature is

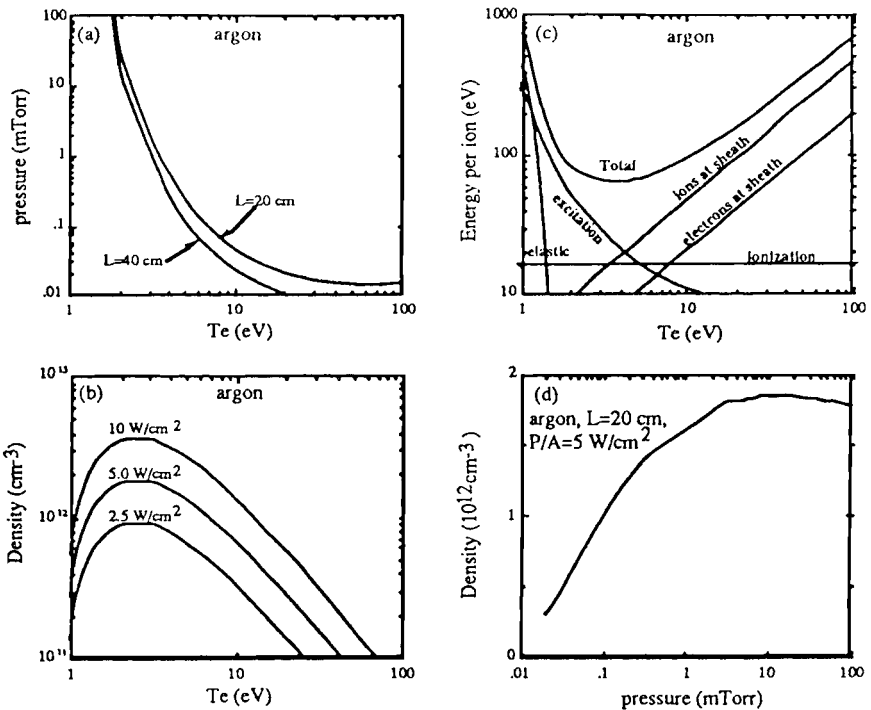
$$\text{Eq. (32)} \quad n_e = P_\mu L / \{ V C_s e [V_p + 2T_e + E_i + E_x \langle \sigma v_e \rangle_x / \langle \sigma v_e \rangle_i + 3T_e m_e / M_i \langle \sigma v_e \rangle_{ei} / \langle \sigma v_e \rangle_{ii}] \}$$

where, for argon, the ionization energy  $E_i \approx 16$  eV and the excitation energy is  $E_x \approx 12$  eV. Plasma density versus  $T_e$  is plotted for argon in Fig. 14b for three power levels. The middle curve with  $P_m L / V \approx 5$  W/cm<sup>2</sup> corresponds to a typical case of 1000 W applied to a 15 cm diameter source. The density is predicted to increase linearly versus power and inversely with source area ( $V/L$ ).

The term enclosed in square brackets in Eq. (32) is the average energy lost per plasma electron-ion pair produced. This term is shown in Fig. 14c along with the relative contributions of the various loss channels for an argon plasma. However, there are additional inelastic processes for realistic processing gases which have not been included in the example here, such as molecular dissociation, rotation, and vibration. The desired result in a processing application is to maximize the energy going into the ions at the sheath which occurs in Fig. 14c at higher electron temperatures. However,  $T_e$  values much exceeding  $\approx 5$ – $8$  eV are undesirable because of increased ion wall sputtering due to the high plasma potential. Application of RF power to the substrate allows the substrate bias to be increased without significantly affecting  $V_p$ .

The global model predicts that density will fall at both low and high pressures as shown in Fig. 14d. This calculated density peaking versus pressure is broader than is typically observed experimentally. A faster density fall-off at high pressure will occur if recombination is included, while local ionization of a significant fraction of the neutrals (*plasma*

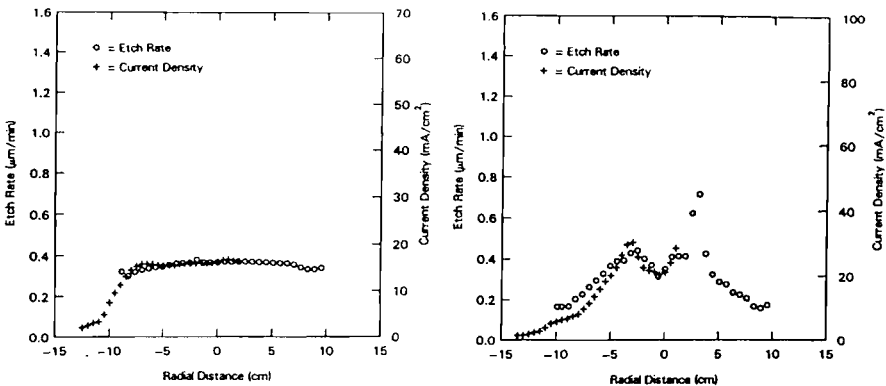
pumping) will increase the density fall-off at low pressures. Correct predictions of these features depend on spatial effects not included in the global model. ECR sources have many other effects which are not included in the global model, among them inhomogeneous electron, ion, and neutral densities, localized microwave power deposition, drifts and non-thermal tails, and various conducting and nonconducting boundary conditions. Thus, two or more dimensional numerical codes are needed to predict quantitative source features such as plasma and process uniformity which depend on the tool geometry. Further refinements and applications of global modeling are discussed by Lieberman et al.<sup>[27]</sup>



**Figure 14.** Global model predictions for an argon ECR plasma: (a) pressure versus electron temperature, (b) density versus electron temperature, (c) energy per ion versus electron temperature, and (d) density versus pressure.

## 2.7 Characterization of ECR Sources for Plasma Processing

Results of ECR plasma processing experiments are often reported as a function of the external tool control parameters such as pressure, flow rate, microwave and RF power, magnetic field, etc. These control parameters vary from tool to tool, making a general understanding of the process difficult. The rationale for plasma characterization studies is that the plasma parameters are closer to the process and are thus the more “universal” process control variables. Huang et al.<sup>[61]</sup> discuss the plasma parameters (reaction parameters) measured at the substrate which are even more specific to the process. These parameters include the ion current density, ion energy, neutral flux, and substrate temperature. For instance, Forster and Holber<sup>[62]</sup> showed a close correlation between the ion current density at the substrate in an ECR tool and the etch rate, and Shapoval, et al.<sup>[24]</sup> showed a similar correlation for deposition. An example of this correlation is shown in Fig. 15 where etch rates of photoresist versus substrate radius scale linearly with the ion current density in an  $O_2$  plasma. Understanding the tool operation from the point of view of the intermediate plasma parameters allows one, in principle, to compare and transfer process results between widely different tools.



**Figure 15.** Radial photoresist etch rates and ion saturation current density (a) flat plasma profile, and (b) hollow plasma profile.<sup>[62]</sup>

### **Ion Current Density, Plasma Density, and Electron Temperature.**

ECR characterization results in the 0.2–10 mtorr pressure range vary from tool to tool but qualitatively show some of the trends versus those predicted by the global model in the last section. Forster and Holber showed that ion saturation current to a downstream substrate in an argon plasma scales almost linearly with microwave power up to 1400 W (Fig. 16). There was a slight threshold in power ( $\approx 200$  W) required to produce the plasma and similar thresholds appear in most reports. Electron temperature increased from 6 to 10 eV while plasma density decreased for pressures decreasing from 1.6 to 0.2 mtorr (Fig. 17). In most reports,  $T_e$  is weakly dependent on the microwave power until the power exceeds a few kW. Ion saturation current measured downstream typically peaks at pressures of  $\approx 1$ –5 mtorr for argon plasmas.<sup>[24][62]–[64]</sup> Figure 18 from Gorbatkin et al.<sup>[42]</sup> shows typical behavior of the downstream density, temperature, and plasma potential versus argon pressure. Density,  $T_e$  and  $V_p$  increase weakly from 25 mtorr down to 4 mtorr, below which density decreases and  $T_e$  and  $V_p$  increase rapidly. Plasma density in the ECR source region tends to peak at an order of magnitude higher pressure and the discharge becomes more localized in the source at higher pressures. The relation between source density and downstream density depends on the gas species. Argon is often used for plasma characterization studies because it is inert and has a mass similar to processing gases like chlorine. Argon typically produces plasma densities a factor of two higher than molecular gases because it has weaker inelastic energy loss mechanisms. Cecchi et al.<sup>[65]</sup> reported an average source density in  $\text{SF}_6$  of  $2.5 \times 10^{12} \text{ cm}^{-3}$  at 1 mtorr and 1200 W which increased with pressure to  $4.5 \times 10^{12} \text{ cm}^{-3}$  at 8 mtorr (Fig. 19). However, the downstream density decreased more rapidly with  $\text{SF}_6$  than with argon as a function of pressure due to electron attachment by the F neutrals. Plasma electron density typically increases with ion mass. In general, ECR plasma parameters depend strongly on pressure and weakly on gas flow rate.

Axial scans of ion current density versus distance made by Samukawa et al.<sup>[66]</sup> with an ion energy analyzer (Fig. 20) showed the  $\text{N}^+$  ion current density peaking near the ECR absorption layer and falling rapidly both downstream and upstream towards the window. Langmuir probe scans made by Popov<sup>[29]</sup> show similar axial density profiles with a drop in density near the window.



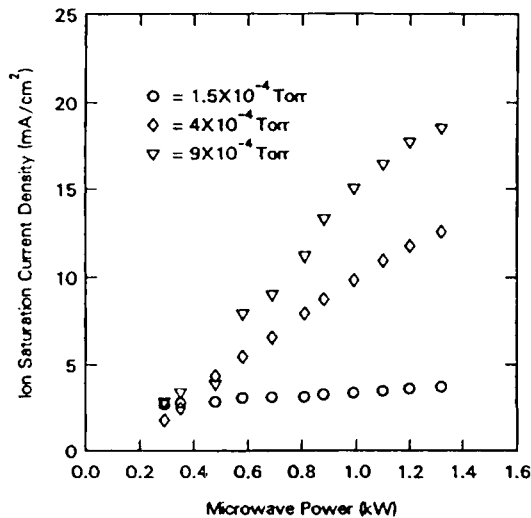


Figure 16. Ion saturation current versus microwave power in argon.<sup>[62]</sup>

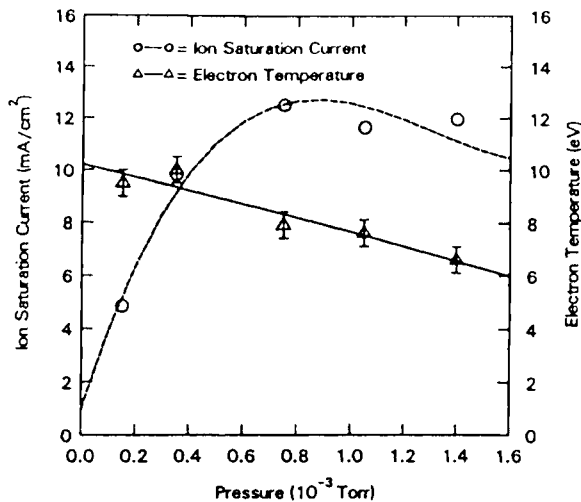
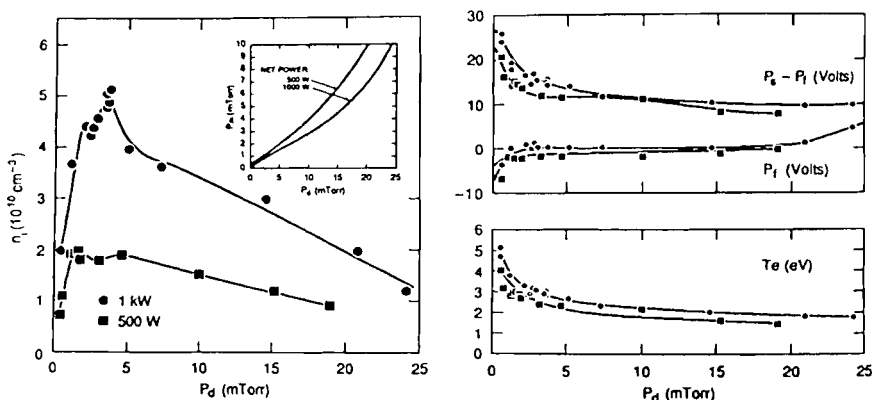


Figure 17. Ion saturation and electron temperature versus pressure at 1300 W in argon.<sup>[62]</sup>

An analysis of experimental results by Popov<sup>[26]</sup> showed that ECR source efficiency increases with source diameter (Fig. 21) for otherwise similar pressure, gas, and power. Plasma density divided by power density,  $N_e/P_s$ , plotted in Fig. 21 scales like the inverse of energy per particle and has not yet clearly saturated for diameters of 25 cm. This indicates that radial losses cannot be neglected in ECR tools even though they have a strong magnetic field. The plasma potential also decreases with diameter while the density decreases slowly with diameter.



**Figure 18.** Plasma density, potential, and electron temperature versus pressure in the downstream ECR chamber.<sup>[42]</sup>

ECR sources produce ion fluxes ranging from 1–50 mA/cm<sup>2</sup> with typical values of  $\Gamma^+ \approx 10$  mA/cm<sup>2</sup> in the downstream mode at 1 mtorr. Etch rates of poly-Si with SF<sub>6</sub> can be  $>1$   $\mu$ /min under these conditions implying a flux of fluorine leaving the surface of  $\Gamma_F \approx 52$  mA/cm<sup>2</sup>. Thus, some of the fluorine used in etching is supplied by neutral atoms and ECR poly-Si etching exhibits an ion enhancement ratio of  $\Gamma_n/\Gamma_+ \approx 2$ –5 between slow neutral and fast ion etchants. Gadgil et al.<sup>[67]</sup> reported ECR poly-Si etching with Cl<sub>2</sub>/O<sub>2</sub> at rates of  $\approx 0.4$   $\mu$ /min with ion fluxes of  $\Gamma_{Cl^+} \approx 30$  mA/cm<sup>2</sup>. This chlorine ion flux exceeds that necessary to account for the etching ( $\Gamma_{Cl} \approx 21$  mA/cm<sup>2</sup>) so it is possible for the etching process to be 100% ion driven

in that case. Parallel plate RIE etchers typically have much more energetic ions and a much higher ratio of neutral to ion flux.

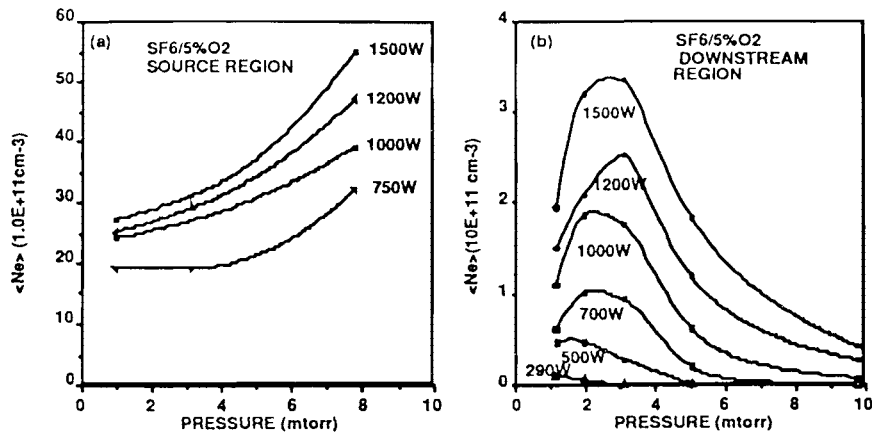


Figure 19. Source and downstream density in SF<sub>6</sub> versus pressure and power.<sup>[65]</sup>

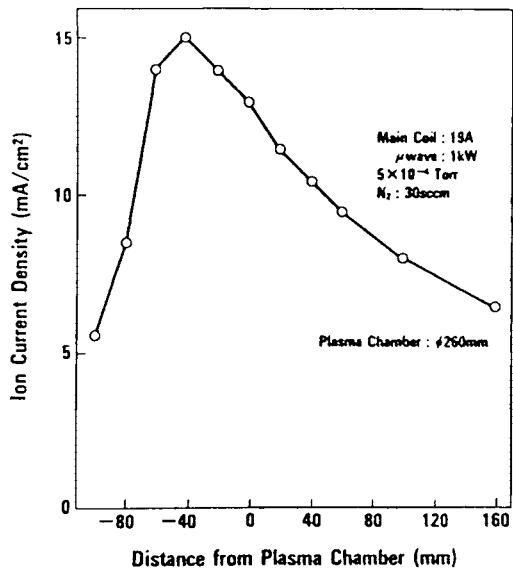
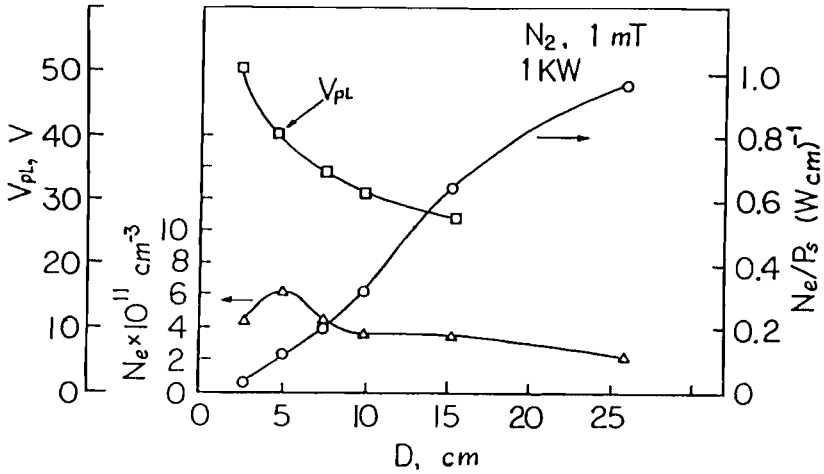


Figure 20. Ion current density measured with an energy analyzer versus axial distance from the ECR source chamber.<sup>[66]</sup>



**Figure 21.** Plasma density, plasma potential, and plasma density per power/area versus source diameter.<sup>[26]</sup>

**Electron Energy Distribution Function in an ECR.** The electron energy distribution function (eedf) is important because it can enhance the ECR source efficiency and may also be a direct or indirect (through UV and x-rays) source of substrate damage. The microwave power will create an electron tail locally at the ECR resonance because the force on resonant electrons from the microwave electric field greatly exceeds the collisional drag force, i.e.,  $m d\bar{v}/dt = -e\bar{E}_\mu - m_e \nu_c \bar{v} \approx -e\bar{E}_\mu$ . The tail produces primary electrons which enhance the source ionization efficiency before transferring their energy to the bulk plasma electrons.

The energy put into electrons passing through the ECR resonance was derived by Kuckes<sup>[7]</sup> for single particle motion in the wave field. The change in perpendicular velocity  $v_\perp$  is

$$\text{Eq. (33)} \quad \Delta v_\perp \approx \eta E_\perp \tau_{\text{res}} \approx \eta E_\perp [2\pi/(\nu_\parallel \eta \nabla_\parallel B)]^{1/2}$$

where in MKS units  $\eta = e/m_e$  is the electron charge to mass ratio,  $E_\perp$  is the microwave electric field in V/m, and  $\tau_{\text{res}}$  is the coherence time of electrons in the microwave field. Equation (33) can be applied to the microwave electric field data plotted in Fig. 11a. The average perpendicular electric field near the resonance is estimated from  $E_\perp \approx (Z_p P/A)^{1/2} \approx 1900$  V/m, where  $P$  is the

microwave power in watts,  $A \approx 0.0177 \text{ m}^2$  is the source area, and the plasma impedance  $Z_p = 377/n_{\parallel} (\Omega)$  was given by Eq. (14). The parallel velocity of resonant electrons is estimated from Eq. (23) as  $v_{\parallel} \approx -c/n_{\parallel}(B/B_{\text{res}} - 1) \approx -2v_{te}$  where  $T_e \approx 6 \text{ eV}$  is taken for the electron temperature. Equations (23) and (33) imply that electrons moving towards the window with an initial velocity  $v_{\parallel 0} \approx 2v_{te}$  and  $v_{\perp 0} = \pm v_{te}$  ( $W_o \approx 24\text{--}30 \text{ eV}$ ) receive an energy kick of  $\Delta W_{\perp} \approx 40 \pm 35 \text{ eV}$  in one pass through the absorption layer, thus putting their final energy near the peak of the ionization cross section around  $80 \text{ eV}$ . The coherence length of an electron in the wave field is  $\approx 3.3 \text{ cm}$ , close to the length over which power is absorbed in Fig. 11a. Fidone<sup>[51]</sup> has derived an expression for the perpendicular electron tail temperature in ECR tools using a quasilinear theory and predicts typical tail temperatures of  $T_{\perp} \approx 4.6T_e$ . Numerical modeling results by Weng et al.<sup>[68]</sup> for an ECR source also predict that a tail with energies of at least  $\approx 50 \text{ eV}$  exists in the source region and is most energetic at lower pressures.

The density of tail electrons,  $n_t$ , can be estimated by equating the energy content of the tail,  $E_t \approx n_t e T_t V_t$ , with the energy dissipated by the microwaves,  $E_{\mu} \approx P_{\mu} \tau_E$ , where  $V_t \approx A v_{\parallel t} \tau_E$  is the volume of plasma where the tail exists,  $A$  is the cross sectional area of the source, and  $t_E$  is the energy loss time of tail electrons. Thus,

$$\text{Eq. (34)} \quad n_t/n_e \approx P_{\mu}/(n_e e T_t A v_{\parallel t}) \approx 10^{-2}$$

where we have estimated  $P_{\mu} \approx 1000 \text{ W}$ ,  $A = 177 \text{ cm}^2$ ,  $v_{\parallel t} \approx 2v_{te}$ ,  $T_e \approx 6 \text{ eV}$ ,  $T_t \approx 4T_e$ , and  $n_e \approx 10^{12} \text{ cm}^{-3}$ . A tail of this magnitude ( $n_t/n_e \approx 1\%$ ) will increase the source ionization efficiency by approximately  $n_t/n_e \langle \sigma v_i \rangle_i / \langle \sigma v_{te} \rangle_i \approx 16\%$  over a purely thermal plasma. This efficiency enhancement varies with source conditions and should be larger for lower  $T_e$  and higher microwave power density. Electron tails of magnitude  $n_t/n_e \approx 1\%$  are difficult to observe with Langmuir probes because one is subtracting a large ion saturation current in order to infer a small current from the high energy electron tail. Langmuir probes are also affected by the microwave electric fields in the region between the window and the ECR layer where the tail is expected to be strongest. An increased ionization efficiency occurs provided the primary electron mean free path is shorter than the source length  $L$ .<sup>[49]</sup> The mean free path for energy loss is determined by ionization, excitation, and electron-electron collisions and is typically  $70 \text{ cm}$  at a millitorr. However, the large momentum transfer cross-sections for electron-neutral and electron-ion collisions will shorten the energy loss distance

in the source by forcing electrons to undergo multiple large angle collisions before escaping.

Suprathermal electron tails have been measured in some ECR plasma processing sources.<sup>[29][42][49]</sup> Quon et al.<sup>[49]</sup> and Tanaka et al.<sup>[42]</sup> observed increasing electron temperatures and suprathermal electron tails at low pressure and high power. Nearly Maxwellian electron distributions over a wide range of ECR parameters have also been reported<sup>[69–71]</sup> in both the downstream and the source region. There have also been reports<sup>[72]</sup> of a depleted high energy (Druyvesteyn) distribution. The importance of maintaining a uniform  $\nabla B_{\parallel}$  versus radius at the ECR absorption layer has been noted<sup>[70][73][74]</sup> and may be related to the uniformity of the electron tail production (Eq. 33).

**Plasma Potential in ECR Sources.** The plasma potential in ECR sources increases as the pressure is lowered with  $V_p \approx 20$  eV typical at 1 mtorr.<sup>[69]</sup> Ono et al.<sup>[63]</sup> showed that the plasma potential in the downstream region decreased from  $> 50$  V to  $< 10$  V as a function distance away from the ECR source for pressures below 0.3 mtorr.

Measurements of plasma potential using  $N_2$  and  $CF_4$  in the ECR downstream region<sup>[75]</sup> showed the existence of a presheath which extended 1.5 ion mean-free-paths from the substrate. Ion energy measurements<sup>[76]</sup> also showed that most of the ion energy gain from the  $-\mu\Delta B$  force is washed out by collisions at typical operating pressures ( $\approx 1$  mtorr) in the downstream region.

Rossnagel et al.<sup>[77]</sup> showed that the plasma density in the source decreased and the radial profile of the ion saturation current tended to flatten as the substrate voltage increased in an ECR etch tool. The converse effect of higher electron density with more negative DC substrate bias was also observed.<sup>[78]</sup> The former experiment<sup>[77]</sup> also showed that a large fraction ( $\approx 86\%$ ) of the ion current was lost to the source walls in a tool with a long narrow source region.

Wei et al.<sup>[79]</sup> showed that applying a DC bias to the source chamber walls simply shifted the plasma potential  $V_p$  up and down so that  $V_p$  remained 20 V more positive than the source walls. Deposition rates of diamond increased as the voltage difference between the plasma potential  $V_p$  and the substrate potential  $V_s$  decreased. No deposition was seen for  $V_p - V_s \geq 23$  V while higher quality diamond occurred for  $V_p - V_s \leq 12$  V.

**Ion Energy and Temperature.** The ion energy is important because it affects selectivity and substrate damage in etching applications and film

quality in deposition applications. ECR sources produce low ion energies of  $V_p - V_f \gtrsim 15$  V at the substrate, where  $V_f$  is the substrate floating potential. Holber<sup>[80]</sup> pointed out that there was a narrow window between 20 eV and 40 eV where silicon could be etched rapidly in a Cl plasma without also rapidly etching SiO<sub>2</sub>. Holber showed that the peak energy and energy spread of ions incident on a downstream energy analyzer decreased with increasing pressure with values of 20 eV  $\pm$  2.5 eV (FWHM) typical for a flared magnetic field at 1 mtorr. Lower ion energies (14  $\pm$  4 eV) occurred for the case of a collimated magnetic field due to less plasma potential difference from  $-\mu\nabla B$  acceleration. Parallel plate diode and reactive ion etch (RIE) tools typically produce ion energies of 100's of eV and Si/SiO<sub>2</sub> selectivity is generally not as good as in low pressure tools.

Ion temperature is an important issue for etching with ECR tools because the ion energy is low.<sup>[80]</sup> An ion perpendicular temperature of 1 eV would produce an angular spread of  $>10^\circ$  at the substrate for a sheath voltage of 36 eV, which would produce sloped sidewalls under the mask. Fujiwara et al.<sup>[37]</sup> measured ion energy spreads of up to 5 eV in a conventional ECR which they attributed to plasma drift instabilities. Attempts were made to reduce ion energy spread utilizing permanent magnets surrounding the source chamber so as to stabilize the instabilities. Straighter sidewalls and more uniform etching as a function of feature size were observed in a silicon etch with Cl as the  $T_{i\parallel}$  measured with an energy analyzer at the substrate decreased from 5.3 eV to 1.8 eV. Further studies<sup>[81]</sup> showed that light ion additives such as He in Cl or the H in HBr reduced the energy spread of the Cl and Br etchant ions, while heavier ion additives such as Xe tended to increase the ion energy spread. The angle of undercut of the mask also increased with increasing ion mass suggesting a higher  $T_i$ , but the plasma potential also increased significantly with the concentration of light ion additives. The ratio of  $T_i/(V_p - V_f) \approx \tan^2\theta$ , which determines the angular spread  $\theta$  of ions at the substrate, decreased with increasing pressure for a given  $V_p - V_f$ . This pointed toward higher pressure with light ion additives as a desirable direction for high anisotropy and high selectivity etching.

High resolution Fabry-Perot measurements of the Doppler-broadened Ar<sup>+</sup> were made by McKillop et al.<sup>[82]</sup> to determine the ion energy distribution in an ECR tool. These line integrated measurements sighting down the axis of a divergent field ECR tool indicated average translational ion energies, characterized as an ion temperature, ranging between  $T_{i\parallel} \approx 2.5$ –1.7 eV and neutral temperatures of  $T_n \approx 0.4$ –0.6 eV for pressures of

0.1–1.5 mtorr. Measurements in a more collimated magnetic field reduced  $T_{\parallel}$  to  $\approx 1.2$  eV and a perpendicular downstream sight line reduced the apparent  $T_n$  to  $\approx 0.15$  eV. O'Neill et al.<sup>[83]</sup> reported perpendicular ion energies of  $T_{\perp} \approx 2.5$ –1.3 eV in the downstream region for both Ar and Cl. The observed spread in ion energies was attributed to ion perpendicular acceleration by gradients in the plasma potential. The reported radial fields of 0.5–1.0 V/cm in the bulk of the downstream plasma were sufficient to accelerate ions to significant energies over an ion mean free path of  $\approx 5$  cm/mtorr, and these fields occurred even for fairly uniform plasma densities.

Nakano et al.<sup>[84]</sup> made localized measurements of Ar<sup>+</sup> and Ne velocities using *laser induced fluorescence* (LIF), and showed  $T_i < 0.5$  eV in the ECR source and  $T_i < 0.25$  eV downstream. Directed ion motion at various locations along a sight line could account for the higher  $T_i$  values seen with the Doppler-broadened emission lines. Sadeghi et al.<sup>[85]</sup> also observed a bimodal ion distribution in the downstream region of an ECR with the faster ions created in the source and the slower ions created near the source-process chamber boundary. DenHartog et al.<sup>[86]</sup> measured perpendicular ion temperatures of 0.12–0.25 eV in N<sub>2</sub> from 4 to 0.5 mtorr. Hershkowitz et al.<sup>[87]</sup> confirmed these  $T_i$  measurements with a pitch angle diagnostic probe where the position of a biased collector in a tube was varied to determine  $T_{i\perp}/E_{\parallel}$ . Goeckner et al.<sup>[88]</sup> reported LIF measurements of room temperature ions in a multidipole chamber with a filament source compared to  $T_{i\perp} \approx 0.46$  eV in a downstream ECR chamber.

**Neutral Density and Temperature.** The density of neutral particles is typically one to two orders of magnitude higher than the plasma density in ECR sources. Neutral gas pressure was reported to decrease significantly when a plasma is present. Gorbatkin et al.<sup>[42]</sup> reported a neutral pressure drop of up to 70% in the ECR source chamber for a power of 1400 W and initial pressures of 0.5–6 mtorr which they attributed to ion pumping by the plasma. Their results were consistent with an argon neutral mean free path of  $\ell \approx 10$  cm in a  $T_e \approx 6$  eV,  $n_e \approx 10^{12}$  cm<sup>-3</sup> plasma source. The lower source neutral pressure was consistent with plasma flows with directed ion energies of  $\approx 5$  eV. Rossnagel et al.<sup>[89]</sup> reported neutral pressure decreases in an ECR source of up to  $\approx 86\%$  in an Ar plasma with an initial pressure of 0.7 mtorr and power of 1250 W. The pressure decrease was attributed to neutral gas heating to temperatures of  $T_n \approx 0.2$  eV and not to depletion by ionization. High neutral temperatures would have important implications for enhanced spontaneous etching which could undercut the mask during etching. The downstream neutral pressure decreased only slightly in both experiments.



Sadeghi et al.<sup>[85]</sup> measured neutral temperatures of  $\approx 350$  K in the processing chamber and  $\approx 800$  K in the ECR source with LIF. These neutral temperatures are in line with the ECR chamber wall temperatures. All of these results imply a large decrease in neutral density in the source which should affect the ECR operation significantly (see Fig. 14a).

Suzuki, et al.<sup>[90]</sup> measured the density of neutral fluorine radicals in  $\text{SF}_6$  plasmas and found values increasing with pressure between  $5 \times 10^{11}$  and  $5 \times 10^{13}$  for pressures from 0.2–50 mtorr. Fluorine densities were an order of magnitude lower in  $\text{CF}_4$ .

## 2.8 RF Biasing of the Substrate in ECR Tools

RF biasing is applied to the substrate in ECR tools in order to control the ion energy independently of the ion flux.<sup>[91]</sup> RF power, applied to a capacitively coupled substrate, creates a negative DC voltage on the substrate which accelerates ions across the sheath. RF biasing in ECR tools is often done at 13.56 MHz although frequencies as low as 0.1 MHz have been studied. The plasma impedance of the RF bias circuit is lower and less reactive than for the commonly used parallel plate tool. This is due to the higher plasma density in ECR tools which supplies a higher ion current per voltage across a narrower, more capacitive sheath. RF biasing of the substrate should ideally not affect the other plasma parameters but, in reality, it can have important consequences for the background plasma.

An important issue for both etching and deposition is how the frequency of RF biasing affects the ion energy at the substrate in ECR tools. Holber et al.<sup>[80]</sup> showed an increased spread in ion energy at RF bias frequencies below a few MHz and at higher RF bias voltages as shown in Fig. 22. This was attributed to the fact that ions can be accelerated directly across the sheath at frequencies  $f \lesssim f_{pi}$  while they will see only the average sheath potential for  $f \gtrsim f_{pi}$ .<sup>[92][93]</sup> The effect of a negatively biased chuck on the plasma is also of interest. A 13.56 MHz RF electrode biased at -20 V was shown by Caughman et al.<sup>[78]</sup> to have only a small effect on the background plasma potential while a -20 V DC bias, which draws a net current from the plasma, tended to reduce the plasma potential and raise the density by  $\approx 50\%$ . It was also observed that biasing at 30 MHz required significantly more power to achieve a given DC bias voltage and caused large RF voltage signals on a downstream probe. Samukawa<sup>[94]</sup> showed that a very low substrate bias frequency ( $< 600$  kHz) reduced oxide break-

down damage during etching of MOS integrated circuits. The self-bias of the wafer disappeared below 600 kHz and ions are accelerated directly by the RF voltage at lower frequencies. An additional RF breakdown at the substrate edge occurred at higher frequency which produced a nonuniform plasma and resulted in substrate damage.

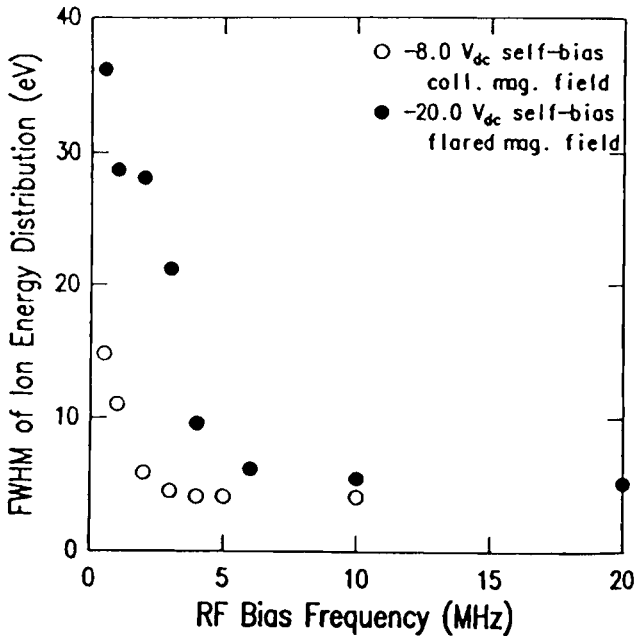
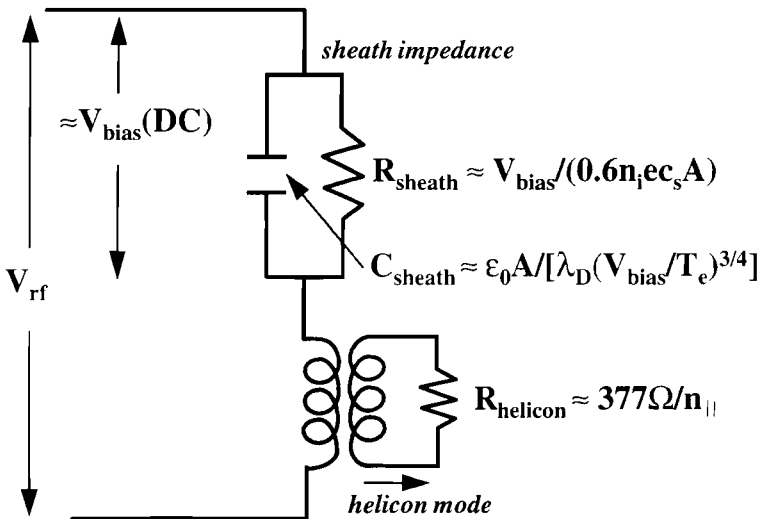


Figure 22. Ion energy spread versus frequency and DC bias.<sup>[80]</sup>

A mapping of ionization and bias voltage versus RF frequency and microwave power was done by Ohtsu, et al.<sup>[95]</sup> for relatively low ECR densities ( $\leq 10^9 \text{ cm}^{-3}$ ). Additional ionization caused by the RF biasing occurred above a certain RF voltage and when the local RF generated density dominated the background microwave density. The RF ionization was attributed to energy absorption in a moving sheath.

Experiments by Cecchi et al.<sup>[96]</sup> at higher density ( $\approx 10^{11} \text{ cm}^{-3}$ ) showed that an RF-biased chuck in an ECR process chamber launches an  $m = 0$

helicon mode into the plasma over the frequency range of  $<1$  to  $>45$  MHz. The RF power in the helicon produces additional ionization locally in front of the substrate which leads to peaking of the density profile and to nonuniform etch rates. The DC bias, RF plasma impedance, and plasma density peaking were mapped over the parameter space of frequency, microwave power ( $\approx$  plasma density), and RF power and explained qualitatively by simple circuit model shown in Fig. 23. The RF voltage to the substrate divides between the sheath impedance and the radiation resistance of the helicon mode. The sheath impedance dominates at low frequency while the helicon impedance,  $Z_h \approx Z_0/n_{||} \approx 10 \Omega$ , dominates at higher frequency. This voltage division accounts for the missing biasing power at higher frequency reported by Caughman.<sup>[78]</sup> The decreasing resistance of the sheath at higher ion density also accounts for the observed reduced DC bias voltage at high microwave powers. Reducing the undesirable effects of the helicon launch from the chuck implies minimizing the RF power relative to microwave power as well as decreasing the RF frequency as low as possible. However, maintaining monoenergetic ions constrains the frequency to  $f \gtrsim f_{pi}$ .



**Figure 23.** RF bias circuit in an ECR tool.

A two-dimensional circuit model for the RF biasing was developed by Namura et al.<sup>[97]</sup> in order to demonstrate how a nonuniform RF current density across the substrate could account for charging damage across thin gate oxides in ECR etch tools. The model postulated a large perpendicular resistivity for the plasma bulk across the magnetic field lines and a smaller parallel resistivity along the field lines in order to account for the nonuniform RF currents drawn to the wafer. Circuit models for the RF biasing of substrates in semiconductor processing tools have been developed primarily for parallel plate tools and these models naturally focused on the capacitive and resistive effects of the substrate and ground-electrode sheaths, while the main body of the plasma is modeled as a low resistance conductor.<sup>[98]</sup> The results discussed above show that the bulk plasma contributes an important term to the RF biasing circuit in ECR sources. A further complication to the RF biasing circuit model in ECR tools is that it is difficult to identify the ground surfaces through which the RF current returns due to the wide variation in densities throughout the tool and the anisotropic nature of the magnetized ECR plasma.

## 2.9 Magnetic Field Control with Coils and Permanent Magnets

The magnetic field configuration has a strong effect on the plasma density and the plasma uniformity in an ECR source. Good density uniformity has been reported when the constant  $|B|$  surfaces were radially uniform at the ECR absorption layer.<sup>[35][56][99]</sup> Figure 24 shows ion current density profiles for a configuration where the  $|B| = 875$  G surface is constant with radius (solid curves) compared to a more typical case where  $|B|$  increases with radius (dashed curves). Lower electron temperatures and plasma potentials were also reported by Samukawa et al.<sup>[70]</sup> for small magnetic field gradients ( $\approx 10$  G/cm) versus larger gradients ( $\approx 40$  G/cm). Other tools utilize larger axial magnetic field gradients ( $\approx 80$  G/cm) with good results as long as  $|B|$  is radially uniform and the ECR layer is located away from the window.

Additional coils can also be utilized to control the downstream plasma profiles. A downstream coil reduces the flaring of the magnetic field which in turn reduces the ion energy impinging on the substrate<sup>[80][100]</sup> and increases the plasma density and ion flux to the wafer. Improved SiO<sub>2</sub> etch rate uniformity with distance from the ECR source was reported by

Fortune-Wiltshire<sup>[101]</sup> at the cost of lower rates for a flared magnetic field. The additional magnetic field of the downstream coil improved the etch rates for a given uniformity provided that the substrate was moved far enough from the ECR source. Other studies by Matsuoka et al.<sup>[100]</sup> and Gorbatkin et al.<sup>[102]</sup> have reported that the downstream coil can improve uniformity as well as increase the density.

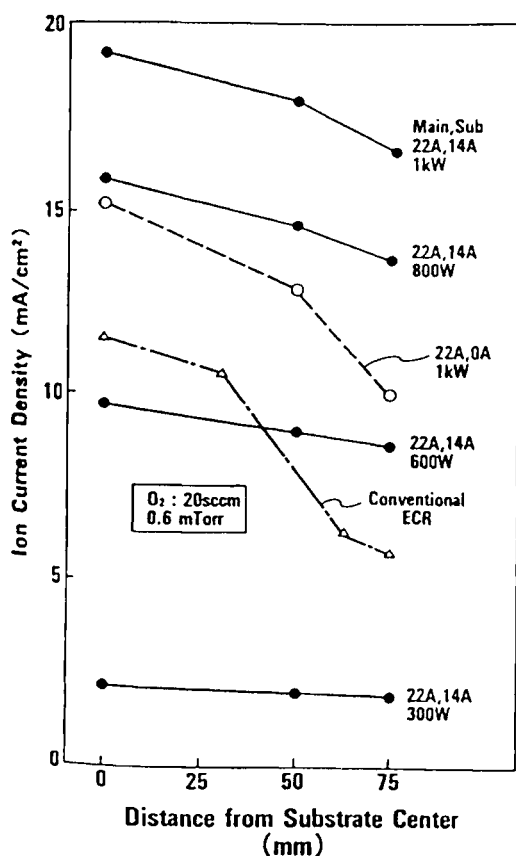


Figure 24. Ion current density versus radius with several magnetic field settings.<sup>[99]</sup>

Another way to significantly improve the density uniformity in ECR tools is to surround the processing chamber by rows of permanent magnets

of alternating polarity creating a multicusp or *magnetic bucket* confinement of the downstream plasma. This leads to a dramatic improvement in plasma uniformity compared with using a diverging magnetic field alone. The advantage of the multicusp configuration is that it seems to remove much of the influence of the source on the plasma uniformity. Boswell et al.<sup>[103]</sup> showed a large improvement in both radial and axial uniformity of the plasma density in a 30 cm dia. by 40 cm downstream chamber filled by a helicon source for argon and  $\text{SF}_6$  at pressures  $<1$  mtorr. Tsai et al.<sup>[104]</sup> reported uniform ion current densities of  $>10$  mA/cm<sup>2</sup> over a 15 cm diameter in a downstream ECR chamber with the addition of a magnetic bucket compared to a peaked profile ranging from  $\approx 7$ –10 mA/cm<sup>2</sup> without the bucket. Mantei et al.<sup>[105]</sup> showed that uniform radial profiles were maintained over axial distances of 5–23 cm from the ECR window with a multipole bucket in the downstream ECR process chamber. Further refinements reported by Gadgil et al.<sup>[106]</sup> with a multipole bucket filled by a single coil ECR source demonstrated radial plasma uniformity in a downstream chamber of 1% ( $1\sigma$ ) over a 20 cm diameter. The radial etch uniformity was 2% over a 15 cm wafer with a high silicon etch rate and high selectivity to  $\text{SiO}_2$  using Cl. Figure 25 shows profiles of ion current density in a downstream ECR for cases with and without a multipole bucket. The axial decrease of ion density, electron temperature, and plasma potential in the downstream chamber is much slower with a magnetic bucket.

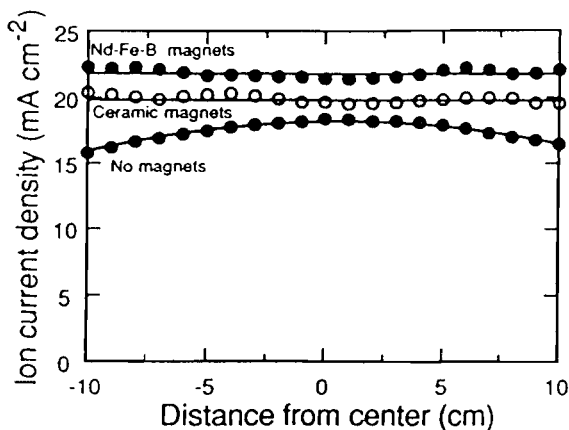


Figure 25. Radial plasma density profiles with and without a magnetic bucket.<sup>[106]</sup>

### 3.0 SPECIAL ECR CONFIGURATIONS AND APPLICATIONS

Many interesting studies have been carried out where the standard ECR source design has been modified in order to improve some aspect of the processing or technical capability of the source. Some of these modified configurations are discussed in this section with no attempt to be all-inclusive.

#### 3.1 Processing in the ECR Source Region

An important innovation for divergent field ECR sources was to do the processing near the ECR resonance layer in the source chamber rather than in a downstream process chamber.<sup>[45][56][57]</sup> These are sometimes referred to as *closely coupled sources*. Among the advantages resulting from this configuration are that the plasma density (i.e., process rate) is higher, the tool is physically more compact, the source is more efficient due to lower ion losses to the source walls, there is less ion energy and ion energy spread from the axial plasma potential difference, and the substrate can be located within a neutral radical mean free path of the ECR resonance layer which increases surface reaction rates. A disadvantage of the closely coupled source is that it requires a larger diameter region of high magnetic field and more precise control of the magnetic field.

Samukawa et al.<sup>[57]</sup> reported achieving a radially uniform plasma and uniform etching in the ECR source region using the closely coupled configuration shown in Fig. 26. The key features cited for creating uniform ion current density were a larger, 17-cm diameter quartz window and subcoils to adjust the magnetic field magnitude to be radially uniform at the ECR location. The ECR resonance layer was also located  $\approx 10$  cm from the rectangular waveguide window. Highly selective etching of n-doped poly-Si over SiO<sub>2</sub> with Cl was reported<sup>[107]</sup> with this closely coupled ECR source.

Fukuda et al.<sup>[45][56]</sup> showed deposition and etch rates a factor of two higher for a substrate in the close position (within  $\approx 10$  cm of the ECR resonance) compared to locations  $> 20$  cm downstream. The quality of deposited Si, W, and AlN films was significantly better than for the downstream substrate location. The uniformity of the etching and deposition was also good. The microwaves are coupled through a conical quartz bell jar and the ECR position was located more than  $> 10$  cm from the bell jar.

Another innovation by Nishimura et al.<sup>[46]</sup> reportedly improved plasma uniformity in closely coupled ECR sources by operating with a smaller inside diameter coil above the window. The small ID coil enhances the magnetic field flaring and thus increases the spreading of the microwaves entering the plasma chamber. An additional feature of Nishimura's design which helped spread the microwave power was a concave quartz window for introducing the microwaves. The concave window gives the waves a less peaked Poynting flux as well as a larger initial  $k_{\perp}$  which allows them to propagate at a larger initial angle to the magnetic field.

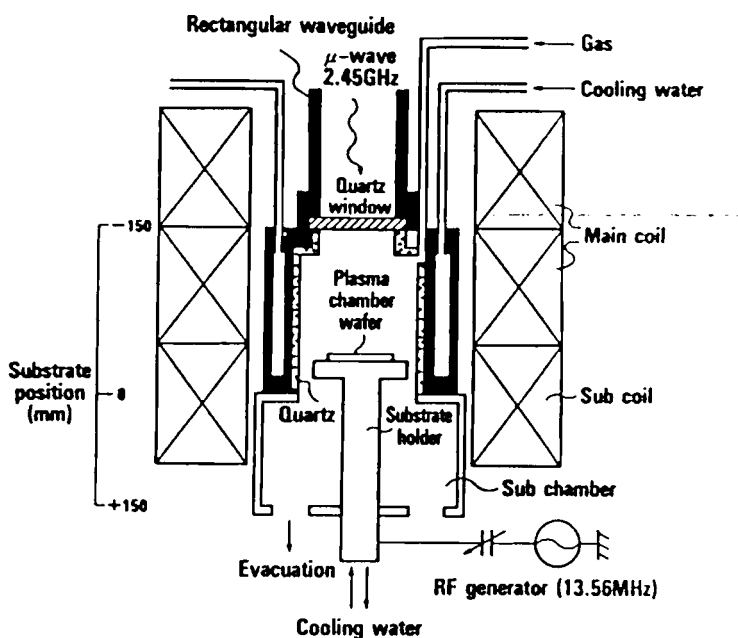


Figure 26. Closely coupled ECR etch tool.<sup>[57]</sup>

### 3.2 High Flow Rate Etching in ECR Sources

The residence time of a gas molecule in a vacuum system is  $\tau = 0.079pV/Q$  (sec), where  $p$  is the pressure in mtorr,  $V$  is the volume in liters, and  $Q$  is the gas flow in sccm. The residence time in most ECR etch tools is typically  $\tau \sim 1$  second and is limited to  $\tau = V/S$  where the pumping speed  $S$  is in liter/sec. An etching gas molecule (F, Cl, etc.) at room temperature will



collide with the walls hundreds of times before being pumped out of a typical ECR tool while it has a high probability of reacting with a silicon surface the first time it hits. Thus, most of the incident species hitting the surface are recycled reaction products. Tsujimoto et al.<sup>[108]</sup> recognized this effect and designed an ECR reactor with high speed (2500 l/s) pumping, a high conductance reaction vessel, and a highly uniform gas flow to the substrate. The high flow rate configuration allows optimum ECR operation at lower pressures than standard ECR tools (0.1–1 mtorr versus 1–10 mtorr). The high flow rate reactor improved the etch rate of crystalline silicon with Cl by an order of magnitude in the low pressure range of 0.1–1 mtorr with peak rates of  $\approx 1.2 \mu\text{m/min}$  at 0.5 mtorr. Improved rates for Al etching with Cl were also cited. This configuration has the additional advantage of reducing ion collisions in the sheath which improved the etching of high aspect ratio features.

### 3.3 Very High Density ECR Sources

Equation 1 implies that there is no upper limit on density at which an ECR source can operate. Although ECR sources for processing typically operate with  $n_e \approx 10^{12} \text{ cm}^{-3}$ , there have been reports where much higher densities have been achieved. Tanaka et al.<sup>[141]</sup> have reported densities of up to  $3 \times 10^{13} \text{ cm}^{-3}$  ( $n_e/n_c \approx 400$ ) using 5 kW of power in a 16 cm diameter system. That particular system utilized a flat magnetic field over an axial distance of  $\approx 70 \text{ cm}$  with  $B/B_{res}$  varied from 0.9–1.7. The maximum density depended on pressure, power, and magnetic field. Two distinct modes were observed depending on those parameters. Density increased approximately linearly with power and pressure in one mode. Above a certain power or below a certain pressure, a second mode began where density saturated and temperature increased approximately linearly with power. The discussion of Sec. 2.6 suggests that this mode transition may occur when the neutral density in the source region becomes depleted enough that operation shifts down and to the right of the curve in Fig. 14a resulting in a  $T_e$  increase.<sup>[58]</sup> This is another argument for closely coupled sources where the neutral density is less likely to become depleted and the density versus power behavior is expected to remain linear.

Higher density can also be achieved at lower power by reducing the plasma size. Reduced plasma size can be achieved in ECR sources by noting that the cutoff wavelength is much smaller in dielectrically loaded waveguides (Eq. 4) so that a dielectric filled window can be tapered down to

small size. Torii et al.<sup>[18]</sup> have produced a 2 cm diameter oxygen ion beam using an ECR source with an  $O^+$  flux of up to 160 mA/cm<sup>2</sup>. That group<sup>[109]</sup> utilized dual quarter-wave transformers (Sec. 2.4) to introduce the microwaves. Plasmas of 5 cm diameter,  $n_e \approx 10^{13}$  cm<sup>-3</sup>,  $T_e \approx 9\text{--}12$  eV were produced<sup>[110]</sup> with  $\approx 200$  kW of 2.45 GHz power. A similar reduced source size was discussed by Popov.<sup>[32]</sup> Helicon sources have been reported<sup>[30][31]</sup> to operate at higher densities ( $n_e \approx 10^{13\text{--}14}$  cm<sup>-3</sup>), especially for small diameter sources operated at higher neutral pressures. However, it is not obvious that either helicon or ECR sources are any more efficient in terms of the number of ions produced per eV of energy delivered when comparable size chambers and pressures are used.

### 3.4 ECR Metal Deposition

A unique property of wave-sustained sources is that the plasma can be created remotely from the ceramic vacuum window. This is especially advantageous for creating plasmas for metal deposition applications. A metal plasma would quickly coat the ceramic window in typical plasma sources and thus prevent the microwave or RF power from entering the plasma. Several authors have reported on ECR metal plasma sources. Metal atoms are fed into the plasma by plasma-sputtering of a biased surface,<sup>[111]–[113]</sup> sputtering of metal from an additional magnetron source,<sup>[114][115]</sup> or evaporation from a filament source.<sup>[116]</sup> In most of these systems, the microwaves were introduced through a window located at high field ( $B > B_{res}$ ) but not connected by magnetic field lines to the plasma column. Holber et al.<sup>[116]</sup> reported an application of ECR sources for copper deposition for advanced interconnect wiring in VLSI circuits. Copper was evaporated into the plasma column from a filament source and was  $\approx 100\%$  ionized in the ECR plasma. The ceramic window was not coated because the ionized copper was confined to the plasma column by the magnetic field lines and copper does not recycle from the vacuum chamber walls. Copper fluxes of  $\approx 10$  mA/cm<sup>2</sup> were produced at the substrate and features with aspect ratios (depth to width) of  $>4$  could be filled with copper at low substrate temperatures. Wave-sustained ECR sources allow the microwaves to propagate and reflect around corners in order to reach the ECR resonance layer and create metal plasmas.

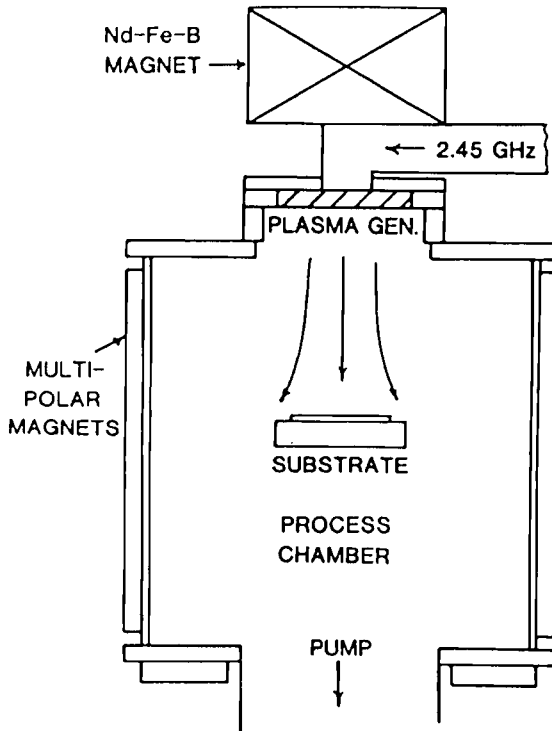
### 3.5 ECR Source Designs With All Permanent Magnets

One drawback of conventional divergent field ECR tools is the use of electromagnets with their associated coils, power supplies, power, and cooling costs. ECR sources have been developed which use permanent magnets to produce the ECR resonance magnetic field. Two embodiments of such ECR sources which are in commercial production are the *distributed ECR* (DECR) tool developed by the CNET group<sup>[117][118]</sup> and the *cavity ECR* tool developed by Asmussen et al.<sup>[118]</sup> DECR sources are discussed in Ch. 8 of this book and ECR cavity sources are discussed in Ch.6 of this book.

Mantei et al.<sup>[120]</sup> and Narai et al.<sup>[121]</sup> have reported ECR sources which have configurations similar to the standard divergent field ECR but which utilize permanent magnets located behind the microwave window to produce a diverging magnetic field. Mantei et al. used a  $15 \times 15 \times 9$  cm, 18 kG block of Nd-Fe-B magnets to generate a magnetic field which falls to 875 G at a distance of 8 cm from the magnet face (Fig. 27). A narrow microwave waveguide is brought in from the side into the narrow space between the magnet face and the window. Plasma radial uniformity was improved by surrounding the process chamber with an additional permanent magnet bucket. Plasma densities of  $>2 \times 10^{11} \text{ cm}^{-3}$  in argon at 500 W and 1 mtorr were achieved. Density and uniformity depended on the distance between the main magnet and the window. Narai et al. utilized two 10 cm dia  $\times$  3 cm thick Sm-Co magnets located 12.6 cm apart, and placed a narrow coupling waveguide and the process chamber between the two. Densities of up to  $10^{12} \text{ cm}^{-3}$  were achieved in  $\text{NF}_3$  at 1 mtorr. Plasma uniformity was improved by coupling through a round hole in the narrow waveguide while higher density was achieved by coupling through slots in the waveguide. The technical improvement of using permanent magnets requires some sacrifice of operational flexibility in adjusting the magnetic fields which makes achieving plasma uniformity more difficult. However, these sources may be ideal for applications where electromagnets are inconvenient or impossible.

An ECR source utilizing permanent magnets inside a grill structure across the ceramic window of a waveguide coupler has also been demonstrated.<sup>[121]</sup> The magnetic field from the permanent magnets is oriented parallel to the window and perpendicular to the microwave electric (E) field.

The plane wave equivalent of this E-field orientation is the extraordinary mode as opposed to the whistler launch parallel to the magnetic field in the standard ECR configuration. The extraordinary mode is also cutoff at  $n_e = n_L$  (Fig. 3). In spite of this, this source can achieve densities well above  $n_c$ . This source can in principle be made arbitrarily large by adding more modules.



**Figure 27.** Permanent magnet design for divergent field ECR source.<sup>[118]</sup>

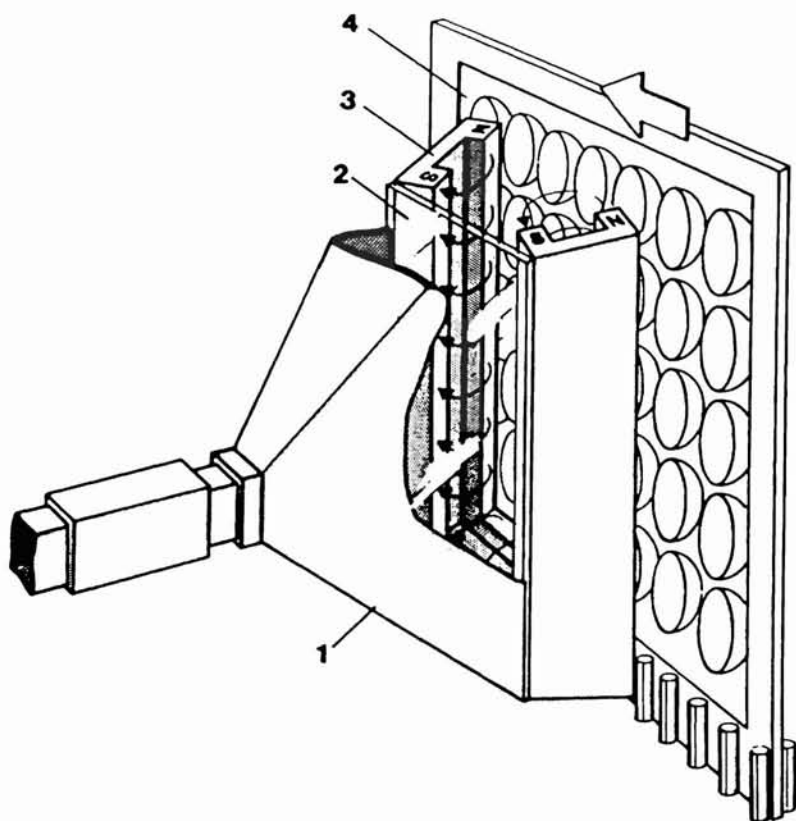
### 3.6 ECR Inline Source for Large Area Coating

Geisler et al.<sup>[122][123]</sup> described an *elongated-mirror-geometry* (EMG) ECR source which is a system suitable for industrial scale deposition onto large surface areas. The EMG source shown in Fig. 28 consists of two long, parallel straight sections which are permanent magnets with a U-shaped cross section. The ends of the “U” are the north and south poles of the permanent magnets and the magnetic field extends into the region between and beyond the ends. The two parallel straight sections are joined top and bottom to make one continuous loop which has the advantage of closing the current loop caused by the  $v_{dx} \approx -\mu \nabla B \times B / e B^2$  drift of electrons perpendicular to the field lines. An elongated microwave horn is fed into the interior of the loop which is under vacuum. The microwaves in this system are launched from the low magnetic field side of the ECR resonance and so must tunnel through a cutoff layer. This does not prevent operation of the source at overdense conditions ( $n_e > n_c$ ). Densities of  $n_e \approx 3n_c$  ( $2.2 \times 10^{11} \text{ cm}^{-3}$ ) near the magnets and  $n_e \approx 1.5n_c$  at a position 17 cm from the magnets were reported. The EMG source is able to achieve deposition rates uniform to within  $\pm 5\%$  over a 3 m length and  $\text{SiO}_2$  deposition rates of up to  $70 \text{ \AA/s}$  were achieved with 2.5 kW of 2.45 GHz power. A 3 m wide belt containing parts or a continuous web is passed in front of the source for coating.

### 3.7 ECR Sources at Widely Different Frequencies and Pressures

The question as to whether there is an optimum frequency for efficiently generating plasmas for processing has been discussed by Moisan et al.<sup>[124]</sup> Low pressure ECR sources operate in a regime where the fractional ionization is relatively high ( $\gtrsim 10^{-3}$ ) and the electron energy distribution is typically a Maxwellian. The frequency will not affect the shape of the high energy tail of the eedf and thus should not matter as far as the efficiency of the ECR source is concerned. The condition  $f \approx f_{ce} \gg \nu_c / 2\pi$  suggests that electrons at frequencies much lower than 2.45 GHz will still undergo cyclotron motion and thus could be used in ECR sources. Lower frequency would have the advantage that lower magnetic fields could be used to produce the ECR resonance making larger volume sources easier. Coil power requirements scale as  $P \approx B^2 \approx f^2$  so an order of magnitude reduction in frequency would reduce the DC power and coil cooling requirements by two orders of magnitude and virtually eliminate them as a concern. Plasma uniformity may also improve with a lower magnetic field due to increased

radial diffusion. Some obstacles of using lower frequency are that cheap RF sources are not presently available between 13.56 MHz and 2.45 GHz, and antenna structures rather than waveguides would have to be designed for frequencies  $\lesssim 0.9$  GHz.



**Figure 28.** Plan view of EMG ECR source for large area processing, showing (1) horn antenna, (2) microwave window, (3) magnetic racetrack system, and (4) substrate holder with discrete substrates.<sup>[122]</sup>

Oda et al.<sup>[125][126]</sup> have implemented an ECR source at a frequency of 144 MHz for amorphous silicon deposition. The center conductor of a 3.5

cm dia. tube was driven with the 144 MHz RF, and evidence for increased plasma production was shown for magnetic fields in the neighborhood of  $B_{res} \approx 51$  G. Effects which might be attributed to the ECR resonance at fields below  $\approx 30$  G have been seen by Yoshida et al.<sup>[127]</sup> in inductively coupled plasmas at 13.56 MHz. The industrial frequency of 915 MHz was studied by Asmussen et al. for cavity ECR sources. Popov et al.<sup>[128]</sup> have utilized 2.45 GHz at the second harmonic to reduce the required ECR magnetic field. A transition to an overdense plasma ( $n_e > n_c$ ) was created when the field at the window was about 437 G, and the plasma could then be maintained over the range of 200–900 G. Absorption was attributed to the upper hybrid resonance occurring near the window.

A 60 GHz, 100 kW ECR system was reported by Arato et al.<sup>[129]</sup> which produced high density ( $10^{12}$  cm<sup>-3</sup>), high temperature plasmas which may be useful as a UV/x-ray source or multicharged ion source. The problem with frequencies higher than 2.45 GHz is that both the microwave sources and the magnetic fields become prohibitively expensive for processing applications.

ECR sources for processing are generally used only in the low mtorr pressure range because of the necessity of having  $\omega_{ce} \gg \nu_c$  in order to have cyclotron motion. Mantei et al.<sup>[130]</sup> used an ECR source for diamond deposition where he showed that there is improved plasma production for pressures as high as 20 torr, although the condition  $\omega \approx \nu_c$  occurs at  $\approx 5$  torr. Hooke et al.<sup>[131]</sup> showed that waves should penetrate into a collisional plasma with  $\nu_c/\omega \gg 1$ , provided that  $\nu_c/\omega_{ce} < 1$ .

### 3.8 Lisitano Coil Coupling

The Lisitano coil<sup>[132][133]</sup> is another well known method for producing plasmas around  $f \approx f_{ce}$  but has not received much attention for processing applications. The microwaves are coupled from the outside of the plasma column by a slotted line or a wire helix surrounding the plasma column and in that sense the coupling geometry of the L-coil source is similar to that of helicon sources. Although the plasma operation is more difficult to quantify due to the complicated antenna, the power must propagate along and across the magnetic field lines and damp by the same ECR mechanism used in other ECR tools. Kawai et al.<sup>[134][135]</sup> constructed a large diameter L-coil for carbon deposition where they reported uniform densities of  $1.5 \times 10^{11}$  cm<sup>-3</sup> over a  $> 30$  cm diameter. Fairly uniform magnetic fields of  $B \approx 875$  G were used in that source. The operation of their coil was investigated by looking

at the vacuum microwave fields and the initial density transient but the uniformity properties with a steady state plasma were not modeled. Hollow plasma density profiles have also been reported by Musil<sup>[25]</sup> using high powers applied to a helix surrounding a plasma column. Densities in the latter case were well above critical ( $2.5 < n_e/n_c < 120$ ) with constant magnetic fields of  $B = B_{res}$  and  $B = 2.2B_{res}$ . Potential advantages of the L-coil are the size scaling and microwave coupling from across magnetic field lines. More development is needed for processing applications in order to better understand the antenna, the uniformity properties of the plasma, and issues such as metal sputtering from the antenna.

## 4.0 OPEN ISSUES FOR ECR SOURCES

### 4.1 Operating Modes

ECR sources have often been described as operating in “modes” where transitions in density or density profiles occur which are sensitive to magnetic field, pressure, and microwave power. Several kinds of mode transitions and their causes have been isolated in the literature. A “high/low” mode transition having to do with the LHP cutoff for densities above and below  $n_e \approx 2n_c$  was discussed in Sec. 2.4. A “peaked/flat” transition in density profile having to do with the microwave propagation interacting with the density profile was discussed in Sec. 2.5. Another “high/low” transition in the slope of the density versus power curve was discussed in Sec. 3.3, characterized by a density saturation and  $T_e$  increase at high levels of power per particle. This transition is likely due to ionization of a large fraction of the neutrals. In addition, there are perhaps more subtle modes related to the mutual interaction between the microwave propagation and the plasma. For instance, Sugai et al.<sup>[14]</sup> described a snakelike behavior of the microwave power flow as it bounced radially back and forth in a long plasma column with the number and length of the bounces depending on density and magnetic field. This could easily give rise to complicated mode behavior in ECR sources and may be an argument for limiting the distance that the microwaves propagate.

An important way to control some mode transitions has recently been identified. It was shown that the long time behavior of certain density profile modes could be stabilized by controlling the neutral density in the



ECR source. Neutral pressure was originally thought to be one of the most important plasma control knobs. Jarmyk et al.<sup>[136]</sup> pointed out that the neutral density will decrease at a constant pressure as the tool walls heat up and that the neutral density was a more important parameter than pressure from the viewpoint of the plasma behavior. By controlling the neutral density through the pumping speed, they were able to stabilize a mode jump indefinitely which had previously occurred over a period of minutes or hours after the start of operation in their system. Aydil et al.<sup>[137]</sup> described modes having two different density levels but having similar profiles. Such mode transitions do not fit the simple peaked/flat classification described earlier which were based solely on RF propagation effects. Aydil, et al. described hysteresis effects for these mode transitions and suggested that the neutral density distribution in the reactor may play a key role.

Time-oscillating mode jumps between the peaked and flat modes discussed in Sec. 2.5 have also been observed.<sup>[55]</sup> These produce a visible “flickering” in the light from the source and occur with frequencies in the Hz to kHz range. Oscillation modes occur when the magnetic field, pressure, and power are near the transition between the two profiles. Neutral gas density depletion in the source chamber alternating with plasma density depletion may also account for this rapid mode jumping. Predicting mode transitions is difficult due to their nonlinear nature, the many variables which are not included in the plasma models, and the lack of time-dependent models for ECR sources.

## 4.2 Plasma Induced Substrate Damage in ECR Sources

Gate widths of  $<0.5\ \mu$  and gate oxide thicknesses of  $<100\ \text{\AA}$  are now required for MOS transistors in VLSI integrated circuits making the processing of these ICs more susceptible to various forms of plasma-induced damage. Plasma-induced damage is a wide ranging subject and only some aspects will be discussed here. Some of the known causes of plasma-induced damage include nonuniform current flow to the wafer, high energy radiation, nondirectional ions, sputtering from metallic surfaces, variable processing results due to wall conditions, physical lattice damage from high energy ions or neutrals, and formation of large “dust” particles. The lack of high energy ions and neutrals is one reason that ECR tools have received such interest for IC manufacturing. The RIE damage threshold for silicon

devices was found by Lee<sup>[137]</sup> to be  $\approx 100$  eV while silicon etching can proceed at typical ECR plasma potentials of 15–30 eV. Little has been reported on particle formation in the plasma at pressures  $< 10$  mtorr but particles formed on the tool walls could be an issue.

Charging of oxide surface layers by ions during etching has been attributed as the cause of damage to the gate oxide. Samukawa et al.<sup>[99]</sup> noted that damaged MOS gates, identified by a low breakdown voltage across the oxide layer, occurred toward the outside of the wafer. The damage occurred for a radially nonuniform ion current density in the ECR tool but did not depend on the magnitude of the ion current density as long as it was uniform across the wafer. The radial profiles in Fig. 24 with solid lines produced no damage while the peaked density profiles (dashed lines) produced damaged gates. Similar results were reported by Namura et al.<sup>[97]</sup> and models for the damage process were proposed by Namura<sup>[138]</sup> and Fang and McVittie.<sup>[139][140]</sup> A nonuniform plasma potential will cause a voltage buildup across the thin gate oxides because the silicon substrate below will maintain an equipotential. Nonuniform ion current or RF bias current flows will also cause large voltage drops to appear across the gate oxides in certain radial locations.

A series of experiments by Mizutani et al.<sup>[142]–[144]</sup> used ECR generated neutral particles to demonstrate that charge carriers and UV radiation are primarily responsible for the oxide damage during etching. Breakdown damage of thin insulating layers is not unique to ECR sources and appears to be a question of producing a radially uniform plasma and RF current at the substrate.

Ion velocities striking the substrate must be normal to the surface to produce good pattern fidelity during etching. ECR tools avoid the problem of ion collisions in the sheath by operating at low pressure. However, a potential problem which may occur for low energy etching of submicron features is that ion trajectories can be deflected from their vertical trajectory by charge accumulation on insulators. Nonvertical ion trajectories may cause undercut or bowing of the sidewalls under the etch mask and may be a problem for all low energy plasma sources. This issue has been investigated by Arnold et al.,<sup>[145]</sup> Davis,<sup>[146]</sup> and McVittie.<sup>[147]</sup>

ECR sources generate UV radiation<sup>[143][144][148]</sup> which has been shown by DiMaria et al.<sup>[149]</sup> to cause trapped charges during RIE etching of SiO<sub>2</sub> gate oxides. Chew et al.<sup>[150]</sup> reported that UV is produced throughout

the plasma volume in an ECR source. Increased resistivity in silicon etched with  $\text{SF}_6$  in an ECR with unbiased substrate was found by Washidzu et al.<sup>[151]</sup> to depths of  $\approx 1 \mu$  presumably due to UV. This silicon damage could be removed by annealing but thermal budgets are becoming more limited. Buchanan et al.<sup>[152]</sup> reported that trapped charge damage in thermally grown  $\text{SiO}_2$  was more easily corrected by annealing after ECR etching than after RIE etching.

X-ray spectra showing electron energies up to 10 keV at 1 mtorr pressures were reported by Castagno, et al.<sup>[153]</sup> from ECR sources. The primary factor affecting x-ray production is neutral pressure, and pressures above a few mtorr are sufficient to eliminate most x-rays while lower pressures result in increasing fluxes and energies. Chew et al.<sup>[154]</sup> reported that x-ray emission tends to be localized from spots where the plasma at the ECR resonance maps along field lines into the wall. It was shown earlier (Sec. 2.7) that electrons are expected to typically get an average energy kick of  $\Delta W_1 \approx 50$  eV while passing through the ECR resonance. Multiple passes through the ECR resonance due to a magnetic mirror have been known to produce suprathermal electrons. Other mechanisms exist for high energy tail production without a mirror. For instance, tail electrons may pitch angle scatter to higher  $v_{\parallel}$  just after they pass through the ECR layer so as to maintain the Doppler-shifted resonance condition for a longer time.

Cecchi et al.<sup>[65]</sup> reported on highly variable etch results which can occur in plasma etching with otherwise similar base pressures and tool parameters if care is not taken with the wall conditions. The effect on the plasma of changes in the process at the substrate is, in fact, utilized for process endpoint detection. Since the substrate surface is generally only a small fraction of the inside surface of the tool and since neutral radicals can reach all inside surfaces, it is not surprising that the surface chemistry of the other tool surfaces can affect the plasma and process behavior.

Various ceramic or anodized aluminum liners have been inserted in ECR sources to eliminate metal contamination from the tool walls. Metallic coatings on the vacuum window have been noted by Gorbatkin et al.<sup>[155]</sup> and Rossnagel et al.<sup>[77]</sup> Gorbatkin et al.<sup>[42]</sup> report that this can significantly reduced the microwave penetration through the window necessitating a hydrogen discharge cleaning procedure to remove the deposits. Great care is taken to avoid metal contamination in production tools.

### 4.3 Plasma Simulation and Modeling of ECR Sources

The expense and difficulty in building and evaluating ECR or other plasma processing tools makes tool simulation an important issue. One would like a modeling tool which could evaluate various aspects of tool operation and design. Models for ECR plasma sources are still in the research stage. No commercial software applications exist for plasma source design analogous to engineering software for circuit design, field analysis, etc. A first-principles model of all aspects of the plasma seems to be out of the question due to the wide variation in time scales which exist in an ECR tool. Some of the important processes which occur in an ECR are electron cyclotron motion on a time scale of  $\tau \approx 10^{-10}$  sec, plasma transport times of  $\tau \approx 10^{-5}$  sec, gas residence times of  $\tau \approx 10^{-1} - 10^0$  sec, a complete processing step for times of  $\tau \approx 10^1 - 10^2$  sec, vessel heating over times of  $\tau \approx 10^2 - 10^3$  sec, and chemical composition changes of the vessel walls for times of  $\tau \approx 10^0 - 10^5$  sec. In addition, the models need to solve for the plasma properties over 2–3 spatial, 2–3 velocity space dimensions, and perhaps over time-dependent features as well. One approach has been to develop hybrid models which solve fluid equations for steady state plasma properties over longer time scales using the results of kinetic equations to determine non-equilibrium plasma properties. The basic model is usually an expansion of the global modeling approach described in Sec. 2.6 to include two-dimensional spatial variations, conducting and insulating boundaries, magnetic field effects, and neutral particle and ion transport. Several ECR modeling approaches are discussed below.

Hussein and Emmert<sup>[156]</sup> presented a one-dimensional (axial) ECR model where the ion trajectories, with charge exchange and elastic collisions with neutrals, were followed with a Monte Carlo calculation and the electrons were assumed to satisfy the Boltzmann relation. Solutions gave the plasma potential, ion energy, and ion temperature as a function of position, pressure, and magnetic field in a downstream ECR. Graves et al.<sup>[157]–[158]</sup> used similar assumptions for the electrons and ions in a two dimensional ( $\hat{r}$  &  $\hat{z}$ ) steady-state model for the ECR plasma. The continuity and energy balance equations were solved iteratively for a steady state solution on the time scale of ion losses ( $\approx 10^{-4}$  sec) with sheaths included by an analytical model. That code calculates steady state plasma parameters such as the density profiles, potential profiles, temperature, and ion current profiles in response to tool geometry, externally set control parameters, and ad-hoc profiles for microwave power deposition. Neutral particle transport

has been added. Porteous et al.<sup>[159]</sup> has included the transport of neutral gas in a helicon source and that model should also be applicable to an ECR.

Kushner et al. have reported several types of hybrid numerical models which were used to explore different aspects of ECR sources. Weng and Kushner<sup>[68]</sup> have calculated the electron energy distribution function in an ECR source using a model where the bulk electrons are treated as a fluid and the tail electrons are followed in a Monte Carlo simulation. Tail electrons were allowed to interact directly with an incoming wave field approaching the ECR layer, as well as interact with the bulk electron, ion and neutral fluids. The time steps were  $\lesssim 10^{-11}$  sec. Model results predicted  $T_e$  values of  $\approx 9$  eV in the source at 1 mtorr falling to  $\approx 3$  eV more than 15 cm downstream with both source and downstream  $T_e$  varying inversely with pressure to some degree. Electron tails extending to energies  $\gtrsim 50$  eV were predicted near the ECR absorption zone. The model also predicted ionization rates and plasma potential versus axial distance, and predicted substantial ionization downstream for pressures  $< 1$  mtorr. Hartig et al.<sup>[160]</sup> used Monte Carlo orbit calculations of the neutral excited states and radicals to model the plasma chemistry in downstream ECR tools. The ions and electrons were treated as a fluid.

## 5.0 SUMMARY

ECR plasma sources have proven to be very useful for plasma processing applications, primarily by exploiting their ability to operate at low pressure and high density in a wave-supported electrodeless mode. The literature of the past thirty-five years has explained many aspects of ECR-source operation and serves to guide the practitioners' intuition and experimentation. Many unanswered questions also remain, especially when integrating various aspects of the ECR-plasma-source behavior with the overall operation of the processing tool.

## ACKNOWLEDGMENTS

I thank my colleagues J. L. Cecchi, C. W. Cheah, Y. C. Huang, R. L. Jarecki, and C. Zuiker for many helpful discussions and the SRC/SEMATECH for research support.

## REFERENCES

1. Miller, D. B., *IEEE Trans. Microwave Theory and Tech.*, MTT-14:162 (1966)
2. Kosmahl, H., Miller, D., and Bethke, G., *J. Appl. Phys.*, 38:4576 (1967)
3. Consoli, T., and Hall, R. B., *Fusion Nucleaire*, 3:237 (1963)
4. Bardet, R., Consoli, T., and Geller, R., *Phys. Letters*, 10:67(1964)
5. Dandl, R. A., England, A. C., Ard, W. B., et al., *Nuclear Fusion*, 4:344 (1964)
6. Levine, A. M., Kuckes, A. F., and Motley, R. W., *Journal of Applied Physics*, 38:4435(1967)
7. Kuckes, A., *Plasma Physics*, 10:367 (1968)
8. Musil J., and Zacek, F., *Plas. Phys.*, 13:471(1971)
9. Musil, J., Zacek, F., and Budnikov, V., *Czech. J. Phys. B*, 23:736 (1973)
10. McVey, B., and Scharer, J., *Phys. Rev. Lett.*, 31:14 (1973)
11. Tulle, P. A., *Plasma Physics*, 16:933 (1974)
12. Stenzel, R. L., *Phys. Fluids*, 19:859 (1976)
13. Musil, J., Zacek, F. and Schmiedberger, P., *Plasma Physics*, 16:735 (1974)
14. Sugai, H., and Takeda, S., *Phys. Fluids*, 23:194 (1980)
15. Stix, T., *Theory of Plasma Waves*, McGraw Hill (1962)
16. Allis, W., Buchsbaum, S., and Bers, A., *Waves in Anisotropic Plasmas*, MIT Press, Cambridge, MA (1961)
17. Matsuo, S., and Adachi, Y., *Jpn. J. Appl. Phys.*, 21:L4 (1982)
18. Torii, Y., Shimada, M., Watanabe, I., *Nucl. Instr. & Meth. B*, 21:178 (1987)
19. Torii, Y., Shimada, M., and Watanabe, I., *Rev. Sci. Instrum.*, 63:2559 (1992)
20. Holber, W., *Handbook of Ion Beam Processing Technology*, (J. Cuomo, S. Rossmagel, and H. Kaufman, eds.), p. 21, Noyes Publications (1989)
21. Tokiguchi, K., Sakudo, N., and Koike, H., *Rev. Sci. Instrum.*, 57:1526 (1986)
22. Suzuki, K., Okudaira, S., Sakudo, N., and Kanomato, I., *Jpn. J. Appl. Phys.*, 16:1979 (1977)
23. Bardos, L., Longar, G., Stoll, I., Musil, J., and Zacek, F., *J. Phys. D: Appl. Phys.*, 8:L195 (1975)
24. Shapoval, S. Y., Petrashov, V. T., Popov, O. A., Yoder, M. D., Maciel, P., and Lok, C. K. C., *J. Vac. Sci. Technol. A*, 9:3071 (1991)
25. Musil, J., *Vacuum*, 36:161 (1986)

26. Popov, O., *Physics of Thin Films*, Vol. 17, (M. Francombe, and J. Vossen, eds.), Academic Press (1993)
27. Lieberman, M. A., and Gottscho, R. A., *Physics of Thin Films*, Vol. 17, (M. Francombe and J. Vossen, eds.), Academic Press (1993)
28. Musil, J., and Zacek, F., *Plasma Phys.*, 12:17 (1970)
29. Popov, O., *J. Vac. Sci. Technol. A*, 9:711 (1991)
30. Boswell, R., and Henry, D., *Appl. Phys. Lett.*, 47:1095 (1985)
31. Chen, F., *J. Vac. Sci. Technol. A*, 10:1389 (1992)
32. Popov, O. A., *J. Vac. Sci. Technol. A*, 7:894 (1989)
33. Carl, D. A., Williamson, M. C., Lieberman, M. A., and Lichtenberg, A. J., *J. Vac. Sci. Technol. B*, 9:339 (1991)
34. Popov, O. A., *J. Vac. Sci. Technol. A*, 8:2909 (1990)
35. Popov, O. A., and Westner, A. O., *Rev. Sci. Instrum.*, 61:303 (1990)
36. Vegas, A., Calderon, M. A. G., and Bustamante, E. G., *Plasma Physics and Controlled Fusion*, 26:1579 (1984)
37. Fujiwara, N., Shibano, T., Nishioka, K., and Kato, T., *Jpn. J. Appl. Phys.*, 28:2147 (1989)
38. Shindo, H., Iseda, S, et al., *Proc. Dry Process Symp.*, Tokyo, 21 (Oct. 1989)
39. Ragan, G. L., *Microwave Transmission Circuits*, MIT Rad. Lab. Series 9, McGraw Hill (1952)
40. Stevens, J. E., Cecchi, J. L., Huang, Y. C., Jarecki, R. L., *J. Vac. Sci. Technol. A*, 9:696 (1991)
41. Tanaka, M., Nishimoto, R., Higashi, S., Harada, N., Ohi, T., Komori, A., Kawai, Y., *J. Phys. Soc. Jpn*, 60:1600 (1991)
42. Gorbatkin, S., Berry, L., and Roberto, J., *J. Vac. Sci. Technol. A*, 8:2893 (1990)
43. Matsuo, S., Nishimura, H., Kiuchi, M., U.S. patent #5,003,152 (Mar. 26, 1991)
44. Takahashi, C., et al., *Symp. VLSI Tech.*, San Diego, 83 (1988)
45. Fukuda, T., Suzuki, N., Takahashi, S., Mochizuki, Y., Ohue, M., Momma, N., and Sonabe, T., *Jpn. J. Appl. Phys.*, 27:L1962 (1988)
46. Nishimura, H., Kiuchi, M., and Matsuo, S., *Jpn. J. Appl. Phys.*, 32:322 (1993)
47. Smith, R. L., Helliwell, R. A., Yabroff, I. W., *J. Geophys. Res.*, 65:815 (1960)
48. Uman, M., and Hooke, W., *Phys. Fluids*, 12:1072 (1969)
49. Quon, B. H., and Dandl, R. A., *Phys. Fluids B*, 1:2010 (1989)

50. Stevens, J. E., Jarecki, R. L., Huang, Y. C., and Cecchi, J. L., *J. Vac. Sci. Technol. A*, 10:1270 (1992)
51. Fidone, I., *report EUR-CEA-FC*, 1464 (June, 1992)
52. Stix, T., *Waves in Plasmas*, p. 269, AIP (1992)
53. Samukawa, S., Sasaki, M., Suzuki, Y., *J. Vac. Sci. Technol. B*, 8:1192 (1990)
54. Tanaka, S., Hamada, Y., et al., U.S. patent #4,970,435 (Nov. 13, 1990)
55. Stevens, J. E., and Cecchi, J. L., *Jpn. J. Appl. Phys.*, 32:3007 (1993)
56. Fukuda, T., Mochizuki, Y., Momma, N., Takahashi, S., Suzuki, N., Sonabe, T., Chiba, K., Suzuki, K., U.S. patent #4,876,983 (Oct. 31, 1989)
57. Samukawa, S., Mori, S., and Sasaki, M., *J. Vac. Sci. Technol. A*, 9:85 (1991)
58. Dandl, R. A., Guest, G. F., *J. Vac. Sci. Technol. A*, 9:3119 (1991)
59. Rapp, D., and Englander-Golden, P., *J. Chem. Phys.*, 43:1464 (1965)
60. Massey, H., *Atomic and Molecular Collisions*, p. 76, Taylor and Francis Ltd., London (1979)
61. Huang, Y. C., Jarecki, R. L., Stevens, J. E., and Cecchi, J. L., *39th AVS Symp.*, Chicago, PS-ThM5 (Nov. 1992)
62. Forster, J., and Holber, W., *J. Vac. Sci. Technol. A*, 7:899 (1989)
63. Ono, T., Oda, M., Takahashi, C., and Matsuo, S., *J. Vac. Sci. Technol. B*, 4:696 (1986)
64. Samukawa, S., Nakagawa, Y., and Ikeda, K., *Jpn. J. Appl. Phys.*, 30:423 (1991)
65. Cecchi, J. L., Stevens, J. E., Jarecki, R. L., and Huang, Y. C., *J. Vac. Sci. Technol. B*, 9:318 (1991)
66. Samukawa, S., Sasaki, M., and Suzuki, Y., *J. Vac. Sci. Technol. B*, 8:1062 (1990)
67. Gadgil, P., Dane, D., Mantei, T., *J. Vac. Sci. Technol. A*, 10:1303 (1992)
68. Weng, Y., and Kushner, M., *J. Appl. Phys.*, 72:33 (1992)
69. Lee, Y., Heidenreich, J., and Fortuno, G., *J. Vac. Sci. Technol. A*, 7:903 (1989)
70. Samukawa, S., and Nakamura, T., *Jpn. J. Appl. Phys.*, 30:3147 (1991)
71. Shirai, K., Iizuka, T., and Gonda, S., *Jpn. J. Appl. Phys.*, 28:897 (1989)
72. Amemiya, H., Shimizu, K., Kato, S., and Sakamoto, Y., *Jpn. J. Appl. Phys.*, 27:L927 (1988)
73. Samukawa, S., and Nakamura, T., *Jpn. J. Appl. Phys.*, 30:L1330 (1991)
74. Fujiwara, N., Sawai, H., et al., *Jpn. J. Appl. Phys.*, 30:3142 (1991)



75. Meyer, J., Kim, G., Goeckner, M., Hershkowitz, N., *Plasma Sources Sci. Technol.*, 1:147 (1992)
76. Hershkowitz, N., Lam, S., Hussein, M., Persing, H., Hartog, E., *44 th GEC Abstracts*, paper JA-1(1991)
77. Rossnagel, S., Schatz, K., Whitehair, S., Guarneri, C., Ruzic, D., Cuomo, J., *J. Vac. Sci. Technol. A*, 9:702 (1991)
78. Caughman, J. B. O., II, and Holber, W. M., *J. Vac. Sci. Technol. A*, 9:3113 (1991)
79. Wei, J., Kawarada, H., Suzuki, J., Ma, J., Hiraki, A., *Jpn. J. Appl. Phys.*, 30:1279 (1991)
80. Holber, W., and Forster, J., *J. Vac. Sci. Technol. A*, 8:3720 (1990)
81. Fujiwara, N., Sawai, H., Yoneda, M., Nishioka, K., and Abe, H., *Jpn. J. Appl. Phys.*, 29:2223 (1990)
82. McKillop, J. S., Foster, J. C., and Holber, W. M., *Appl. Phys. Lett.*, 55:30 (1989)
83. O'Neill, J., Holber, W., and Caughman, J., *IBM Research Report*, RC16439#73052 (1991)
84. Nakano, T., Sadeghi, N., and Gottscho, R. A., *Appl. Phys. Lett.*, 58:458 (1991)
85. Sadeghi, N., Nakano, T., Trevor, D., and Gottscho, R., *J. Appl. Phys.*, 70:2552 (1991)
86. Den Hartog, E. A., Persing, H., and Woods, R. C., *Appl. Phys. Lett.*, 57:661 (1990)
87. Hershkowitz, N., Lam, S., Hussein, M., Persing, H., Hartog, E., *44 th GEC Abstracts*, DA-5 (1991)
88. Goeckner, M., Goree, J., and Sheridan, T., *J. Vac. Sci. Technol. A*, 9:3178 (1991)
89. Rossnagel, S. M., Whitehair, S. J., Guarneri, C. R., and Cuomo, J. J., *J. Vac. Sci. Technol. A*, 8:3113 (1990)
90. Suzuki, K., Ninomiya, K., and Nishimatsu, S., *Vacuum*, 34:953 (1984)
91. Suzuki, K., Ninomiya, K., Nishimatsu, S., and Okudaira, S., *J. Vac. Sci. Technol. B*, 3:1025 (1985)
92. Coburn, J., and Kay, E., *J. Appl. Phys.*, 43:4965 (1972)
93. Kohler, K., Horne, D., and Coburn, J., *J. Appl. Phys.*, 58:3350 (1985)
94. Samukawa, S., *Jpn. J. Appl. Phys.*, 30:3154 (1991)
95. Ohtsu, Y., Okuno, Y., and Fujita, H., *Jpn. J. App. Phys.*, 32:2873 (1993)
96. Cecchi, J. L., Stevens, J. E., Cheah, C. W., Huang, Y. C., Jarecki, R. L., and Zuiker, C., *40th AVS Symp.*, Orlando, PS-TuA6 (Nov. 1993)

97. Namura, T., Okada, H., Naitoh, Y., Todokoro, Y., and Inoue, M., *Jpn. J. Appl. Phys.*, 29:2251 (1990)
98. Chapman, B., *Glow Discharge Processes*, John Wiley & Sons, New York (1980)
99. Samukawa S., and Nakamura, T., *Jpn. J. Appl. Phys.*, 30:L1902 (1991)
100. Matsuoka, M., and Ono, K., *Appl. Phys. Lett.*, 50:1864 (1987)
101. Fortune-Wiltshire, G., *J. Vac. Sci. Technol. A*, 9:2356 (1991)
102. Gorbatkin, S. M., Berry, L. A., and Swyers, J., *J. Vac. Sci. Technol. A*, 10:1295 (1992)
103. Boswell, R., Perry, A., and Emami, M., *J. Vac. Sci. Technol. A*, 7:3345 (1989)
104. Tsai, C. C., Berry, L. A., et al., *J. Vac. Sci. Technol. A*, 8, 2900 (1990)
105. Mantei, T., and Ryle, T., *J. Vac. Sci. Technol. B*, 9:29 (1991)
106. Gadgil, P., Dane, D., Mantei, T., *J. Vac. Sci. Technol. B*, 11:216 (1993)
107. Samukawa, S., Suzuki, Y., and Sasaki, M., *Appl. Phys. Lett.*, 57:403 (1990)
108. Tsujimoto, K., Kumihashi, T., Kofuji, N., and Tachi, S., *Proc. 14th Dry Process Symp.*, Tokyo, 49 (Oct. 1992)
109. Shimada, M., Watanabe, I., and Torii, Y., *Nucl. Instrum. Methods B*, 39:242 (1989)
110. Shimada, M., Watanabe, I., and Torii, Y., *J. Vac. Sci. Technol. A*, 9:707 (1991)
111. Matsuoka, M., and Ono, K., *Appl. Phys. Lett.*, 54:1645 (1989)
112. Takehiro, S., Yamanaka, N., Shindo, H., Shingubara, S., and Horiike, Y., *Jpn. J. Appl. Phys.*, 30:3657 (1991)
113. Kidd, P., *J. Vac. Sci. Technol. A*, 9:466 (1991)
114. Takahashi, C., Kiuchi, M., Ono, T., and Matsuo, S., *J. Vac. Sci. Technol. A*, 6:2348 (1988)
115. Yoshida, Y., *Appl. Phys. Lett.*, 61:1733 (1992)
116. Holber, W., Logan, J. S., Grabarz, H. J., Yeh, J. T., Caughman, et al., *J. Vac. Sci. Technol. A*, 11:2903 (1993)
117. Pichot, M., Durandet, A., Pelletier, J., Arnal, Y., and Vallier, L., *Rev. Sci. Instrum.*, 59:1072 (1988)
118. Hopwood, J., Dahimene, M., Reinhard, D. K., and Asmussen, J., *J. Vac. Sci. Technol. B*, 6:268 (1988)
119. Mantei, T. D., and Dhole, S., *J. Vac. Sci. Technol. B*, 9:26 (1991)
120. Narai, A., Hashimoto, T., Ichihashi, H., Shindo, H., and Horiike, Y., *Jpn. J. Appl. Phys.*, 30:3159 (1991)

121. Getty, W. D. and Geddes, J. B., *J. Vac. Sci. Technol. B*, 12:408 (1994)
122. Geisler, M., Kieser, J., Rauchle, E., Wilhelm, R., *J. Vac. Sci. Technol. A*, 8:908 (1990)
123. Wilhelm, R., *Plasma Phys. Control. Fusion*, 35:A199 (1993)
124. Moisan, M., Barbeau, C., Claude, R., Ferreira, C., Margot, J., Paraszczak, J., Sá, A., Sauvé, G., and Wertheimer, M., *J. Vac. Sci. Technol. B*, 9:8 (1991)
125. Oda, S., *Plasma Sources Sci. Technol.*, 2:26 (1993)
126. Oda, S., Noda, J., and Matsumura, M., *Jpn. J. Appl. Phys.*, 28:L1860 (1989)
127. Yoshida, A., Kitagawa, M., Setsune, K., and Hirao, T., *Jpn. J. Appl. Phys.*, 27:L1355 (1988)
128. Popov, O., Shapoval, S., and Yoder, M., *Plasma Sources Sci. Technol.*, 1:7 (1992)
129. Arato, Y., Miyake, S., Kishimoto, H., Abe, N., and Kawai, Y., *Jpn. J. Appl. Phys.*, 7:1281 (1988)
130. Mantei, T. D., and Chang, J. J., *J. Vac. Sci. Technol. A*, 10:1423 (1992)
131. Hooke, W., Bozeman, S., Olesik, J., *Proc. Course Industrial Appls. of Plasma Phys.*, Varena, Italy:33 (1992)
132. Lisitano, G., Ellis, R. A., Hooke, W. M., and Stix, T. H., *Rev. Sci. Instrum.*, 39:295 (1968)
133. Lisitano, G., *Proc. Course on Industrial Applications of Plasma Phys.*, Varena, Italy:109 (1992)
134. Kawai, Y., Komori, A., Murata, M., *Proc. Course Industrial Appls. of Plasma Phys.*, Varena, Italy:117 (1992)
135. Yonesu, A., Takeuchi, Y., Komori, A., and Kawai, Y., *Jpn. J. Appl. Phys.*, 27:L1746 (1988)
136. Jarnyk, M., Gregus, J., Aydil, E., Gottscho, R., *Appl. Phys. Lett.*, 62:2040 (1993)
137. Aydil, V., Gregus, J., and Gottscho, R., *J. Vac. Sci. Technol. A*, 11:2883 (1993)
138. Lee, Y., *J. Vac. Sci. Technol. A*, 10:1318(1992)
139. Namura, T., Uchida, H., Todokoro, Y., and Inoue, M., *J. Vac. Sci. Technol. B*, 9:2752, (1991)
140. Fang, S., and McVittie, J., *IEEE Electron Device Lett.*, 13:288 (1992)
141. Fang, S., and McVittie, J., *IEEE Electron Device Lett.*, 13:347 (1992)
142. Mizutani, T., and Nishimatsu, S., *J. Vac. Sci. Technol. A*, 6:1417 (1988)
143. Mizutani, T., and Nishimatsu, S., *J. Vac. Sci. Technol. B*, 7:547, (1989)
144. Mizutani, T., and Yunogami, T., *Jpn. J. Appl. Phys.*, 29:2220 (1990)

145. Arnold, J., and Sawin, H., *J. Appl. Phys.*, 70:5314 (1991)
146. Davis, R., *Appl. Phys. Lett.*, 59:1717 (1991)
147. McVittie, J., abstract, *2nd Workshop on High Density Plasma Sources*, San Francisco, CA, (Aug. 1993)
148. Yapsir, A., Fortuno-Wiltshire, G., Gambino, J., Kastl, R., and Parks, C., *J. Vac. Sci. Technol. A*, 8:2939 (1990)
149. DiMaria, D., Ephrath, L., and Young, D., *J. Appl. Phys.*, 50:4015 (1979)
150. Chew, K. H., Castagna, T. J., Steward, K. L., Mertz, F. F., and Shohet, J. L., *J. Vac. Sci. Technol. A*, 11:1186 (1993)
151. Washidzu, G., Hara, T., Hiyoshi, J., *Jpn. J. Appl. Phys.*, 30:1045 (1991)
152. Buchanan, D. A., and Fortuno-Wiltshire, G., *J. Vac. Sci. Technol. A*, 9:804 (1991)
153. Castagno, T. J., Shohet, J. L., Denton, D., and Hershkowitz, N., *Appl. Phys. Lett.*, 60:2856 (1992)
154. Chew, K. H., and Shohet, J. L., *Proc. of AVS 39th Nat. Symp.*, PS-TuM7:171 (1992)
155. Gorbatkin, S. M., and Berry, L. A., *J. Vac. Sci. Technol. A*, 10:3104 (1992)
156. Hussein, M. A., and Emmert, G. A., *J. Vac. Sci. Technol. A*, 8:2913 (1990)
157. Graves, D. B., Wu, H., Porteous, R. K., *Proc. Dry Process Symp.*, Tokyo, Japan (1992)
158. Porteous, R., and Graves, D., *IEEE Trans. Plas. Sci.*, 19:204 (1991)
159. Porteous, R., and Boswell, R., *45th GEC Abstracts*, paper LB-13, Boston (1992)
160. Hartig, M., and Kushner, M., *Appl. Phys. Lett.*, 62:1595 (1993)